LAB # 9

Yusuf Bozdağ 33840 Lab Section : C

1. Introduction

n this experiment, a discrete multistage BJT amplifier was designed, simulated, built, and tested in order to satisfy specific voltage gain, bandwidth, and impedance requirements. The main objective was to obtain a non-inverting midband voltage gain of approximately 100 V/V while maintaining a lower 3-dB frequency below 20 Hz, an upper 3-dB frequency above 20 kHz, an input impedance greater than 1 MΩ, and an output impedance lower than 100 Ω. The amplifier was also expected to drive a 200 Ω load and produce a 2.5 Vpp output signal at 1 kHz without nonlinear distortion. Since the circuit had to be constructed using discrete BJTs, resistors, and capacitors, the design required careful selection of transistor configurations, biasing conditions, coupling capacitors, and load-driving capability.

The amplifier circuit was based on a four-stage structure using 2N2222 transistors. The first stage was designed as an emitter follower to provide high input impedance and prevent the signal source from being loaded. The second and third stages were common-emitter amplifiers, which supplied most of the voltage gain. Because each common-emitter stage introduces a 180° phase shift, using two of them allowed the total amplifier response to remain non-inverting. The final stage was another emitter follower, used as an output buffer to reduce the output impedance and provide enough current to drive the low 200 Ω load. This stage-by-stage structure allowed the gain, impedance, and bandwidth requirements to be satisfied more reliably than trying to obtain the full gain from a single transistor stage.

The theoretical operation of the circuit depends on the small-signal behavior of BJT amplifier stages. In the common-emitter stages, small variations in base-emitter voltage create amplified collector-current variations, which are converted into voltage changes through the collector resistors. The emitter resistors improve bias stability and linearity, although they also reduce the voltage gain. In contrast, the emitter follower stages have voltage gain close to unity but provide important impedance buffering. Coupling capacitors were used between stages and at the output to block DC current from flowing through the signal source and load while allowing the AC signal to pass. The capacitor values were selected so that the low-frequency cutoff would remain below the required 20 Hz limit.

Before implementing the circuit in the lab, hand calculations and LTspice simulations were used to verify the DC operating points, midband gain, input and output impedances, transient response, and frequency response. This comparison was necessary because practical BJT parameters, especially current gain β, can vary significantly from ideal or simulated values. Therefore, the experiment also aimed to observe how theoretical predictions, simulation results, and real laboratory measurements differ due to nonideal transistor behavior, component tolerances, loading effects, and possible clipping or distortion. Overall, this lab demonstrates how a practical multistage amplifier can be designed by combining gain stages with buffer stages to meet realistic circuit specifications.

2. Pre-Lab Work

In this pre-lab, a discrete multistage BJT amplifier is designed using 2N2222 transistors in order to satisfy the following specifications: a non-inverting midband voltage gain of approximately $100 \pm 15\%$, a lower 3-dB frequency below 20 Hz, an upper 3-dB frequency above 20 kHz, an input impedance higher than 1 MΩ, an output impedance lower than 100 Ω, and an output swing of 2.5 Vpp at 1 kHz without nonlinear distortion. The circuit operates from symmetric supply voltages of $+15$ V and -15 V, and it must not allow DC current to flow through the signal source or through the load.

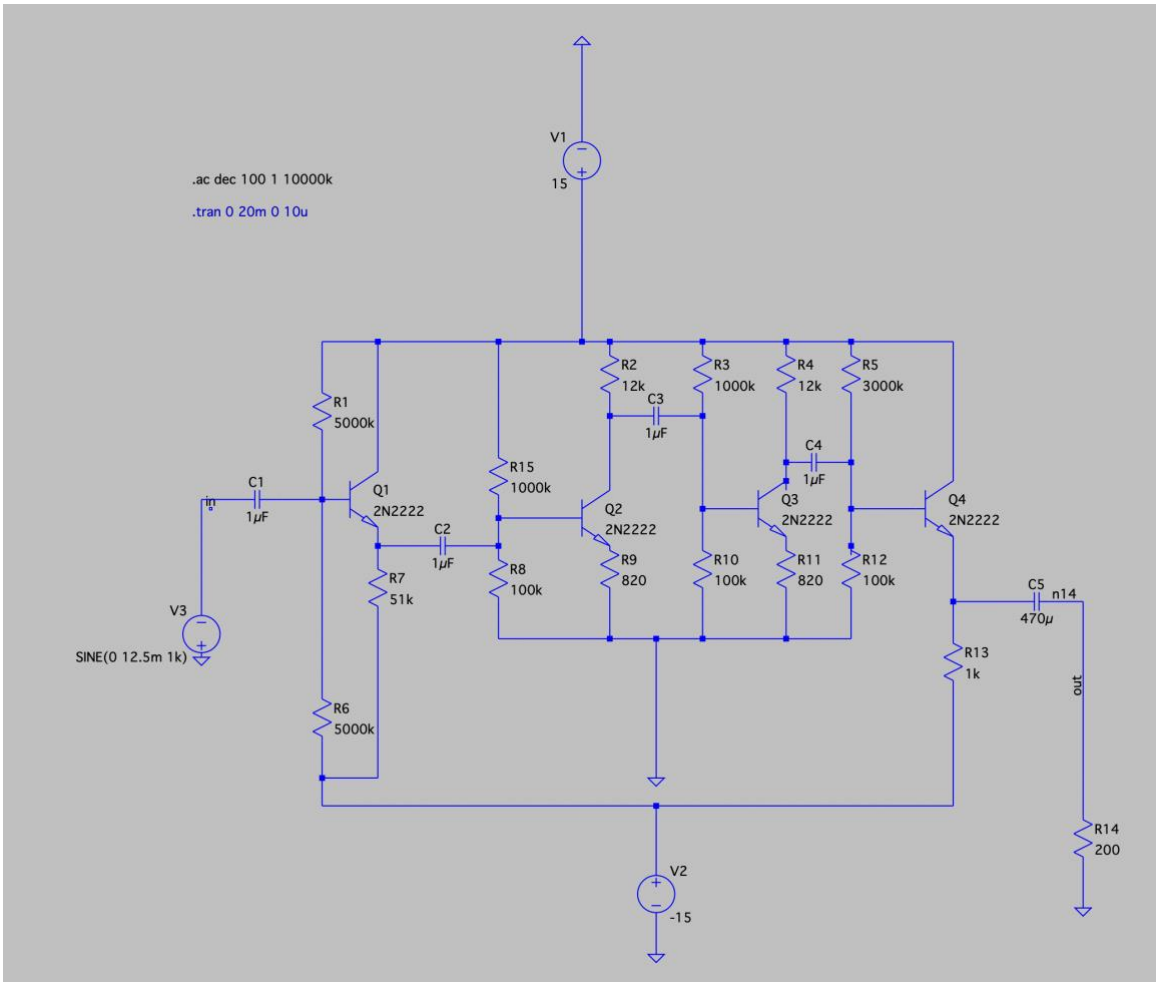


Figure 1. Schematic of Multi-Stage BJT Amp.

The final amplifier consists of four transistor stages. The first transistor, Q_1, is an emitter follower used as the input buffer. Its main purpose is to provide very high input impedance so that the source is not loaded. The second and third transistors, Q_2 and Q_3, are common-emitter voltage-gain stages. These two stages provide almost all of the required voltage amplification. Since each common-emitter stage introduces a phase inversion, using two such stages results in an overall non-inverting response. The last transistor, Q_4, is another emitter follower used as the output buffer. This stage lowers the output

impedance and allows the circuit to drive the $200\ \Omega$ load. All stages are AC-coupled, which ensures that there is no DC current through the signal source or through the load.

The selected component values are

$R_1 = 5$ MΩ, $R_6 = 5$ MΩ, $R_7 = 51$ kΩ, $C_1 = 1\ \mu$F

$R_{15} = 1$ MΩ, $R_8 = 100$ kΩ, $R_2 = 12$ kΩ, $R_9 = 820\ \Omega$, $C_2 = 1\ \mu$F

$R_3 = 1$ MΩ, $R_{10} = 100$ kΩ, $R_4 = 12$ kΩ, $R_{11} = 820\ \Omega$, $C_3 = 1\ \mu$F, $C_4 = 1\ \mu$F

$R_5 = 3$ MΩ, $R_{12} = 100$ kΩ, $R_{13} = 1$ kΩ, $C_5 = 470\ \mu$F, $R_L = 200\ \Omega$

For the hand calculations, a typical value of $\beta \approx 200$ is assumed for the 2N2222 in the intended operating region. In practice, β is not constant, so small differences between hand calculations and LTspice results are expected. However, this assumption gives results that match the simulations quite well.

Design strategy and why each component value was chosen

The first design target was the input impedance. To meet the condition $Z_{in} \geq 1$ MΩ, the input bias divider of Q_1 had to use very large resistor values. That is why R_1 and R_6 were both selected as 5 MΩ. Their parallel combination is 2.5 MΩ, which is already comfortably above 1 MΩ. If smaller bias resistors had been used, the input stage would have loaded the source too much.

The first transistor, Q_1, was chosen as an emitter follower because an emitter follower has high input resistance, near-unity voltage gain, and good buffering action. The emitter resistor $R_7 = 51$ kΩ was selected to keep the first-stage current relatively small while still allowing proper operation. Since this stage does not need to provide large voltage gain, a low current is sufficient.

The second and third stages were designed as common-emitter amplifiers because these are the main voltage-gain stages. The collector resistors R_2 and R_4 were selected as 12 kΩ to obtain substantial gain while still leaving enough collector voltage headroom. The emitter resistors R_9 and R_{11} were selected as $820\ \Omega$, slightly smaller than 1 kΩ, in order to increase the gain somewhat compared to the previous version while still maintaining emitter degeneration for linearity and bias stability. This change is especially important because 2N2222 generally has lower βthan 2N5089, so some compensation was necessary to keep the total gain in the desired range.

The bias dividers R_{15}-R_8 and R_3-R_{10} were both chosen as 1 MΩ and 100 kΩ, respectively. These values place the bases of Q_2 and Q_3 slightly above 1 V, which sets the emitter currents in the desired sub-milliamp range. At the same time, the divider resistances are large enough that they do not excessively load the previous stage.

The output stage had to satisfy the low output impedance requirement and also drive a $200\ \Omega$ load with 2.5 Vpp. For this reason, Q_4 was also implemented as an emitter follower, but biased at a much higher current than the earlier stages. The bias network $R_5 = 3$ MΩ and $R_{12} = 100$ kΩ, together with $R_{13} = 1$ kΩ, sets a sufficiently large emitter current so that the small-signal emitter resistance becomes very small. This is the key reason why the output impedance remains far below 100 Ω.

Finally, the coupling capacitors were chosen according to the low-frequency specification. C_1, C_2, C_3, and C_4 were selected as $1\ \mu$F, which is sufficient because the surrounding resistances are in the tens of kΩ or higher. However, the output capacitor had to work with a very small load resistance of only $200\ \Omega$. For that reason, C_5 was made much larger, $470\ \mu$F, in order to keep the output high-pass pole well below 20 Hz.

DC bias calculations

2.1 Bias of Q_1

The base of Q_1 is biased by R_1 and R_6 connected between $+15$ V and -15 V. The Thevenin equivalent of this divider is

$$V_{TH1} = 0 \text{ V}$$
$$R_{TH1} = R_1 \parallel R_6 = 5\ \text{MΩ} \parallel 5\ \text{MΩ} = 2.5\ \text{MΩ}$$

If V_{B1} is the base voltage, then

$$V_{E1} = V_{B1} - 0.7$$
$$I_{E1} = \frac{V_{E1} + 15}{R_7} = \frac{V_{B1} + 14.3}{51\ \text{kΩ}}$$
$$I_{B1} = \frac{I_{E1}}{\beta + 1}$$

and because the divider is loaded by base current,

$$V_{B1} = V_{TH1} - I_{B1} R_{TH1}$$

Substituting $\beta = 200$,

$$V_{B1} = -\frac{2.5 \times 10^6}{201 \cdot 51 \times 10^3}(V_{B1} + 14.3)$$

Solving gives

$$V_{B1} \approx -2.80 \text{ V}$$

Therefore,

$$V_{E1} \approx -3.50 \text{ V}$$
$$I_{E1} = \frac{-3.50 + 15}{51\ \text{kΩ}} \approx 0.225\ \text{mA}$$

So Q_1 is biased at a low but sufficient current, which is appropriate for an input buffer.

2.2 Bias of Q_2

The bias divider of Q_2 is formed by $R_{15} = 1\ \text{M}\Omega$ and $R_8 = 100\ \text{k}\Omega$, connected between $+15$ V and ground.

$$V_{TH2} = 15 \cdot \frac{100\ \text{k}\Omega}{1\ \text{M}\Omega + 100\ \text{k}\Omega} = 1.364\ \text{V}$$
$$R_{TH2} = 1\ \text{M}\Omega \parallel 100\ \text{k}\Omega = 90.9\ \text{k}\Omega$$

Since the emitter resistor is $R_9 = 820\ \Omega$,

$$V_{E2} = V_{B2} - 0.7$$
$$I_{E2} = \frac{V_{B2} - 0.7}{820}$$
$$I_{B2} = \frac{I_{E2}}{201}$$
$$V_{B2} = V_{TH2} - I_{B2}R_{TH2}$$

Solving gives

$$V_{B2} \approx 1.128\ \text{V}$$
$$V_{E2} \approx 0.428\ \text{V}$$
$$I_{E2} \approx \frac{0.428}{820} \approx 0.522\ \text{mA}$$

Assuming $I_C \approx I_E$,

$$I_{C2} \approx 0.52\ \text{mA}$$

Then the collector voltage is

$$V_{C2} = 15 - I_{C2}R_2 = 15 - (0.522\ \text{mA})(12\ \text{k}\Omega) \approx 8.77\ \text{V}$$

Thus,

$$V_{CE2} = V_{C2} - V_{E2} \approx 8.77 - 0.43 = 8.34\ \text{V}$$

This confirms that Q_2 is safely in the active region.

2.3 Bias of Q_3

Since Q_3 uses the same bias network and the same collector and emitter resistors as Q_2, its DC operating point is almost identical:

$$V_{B3} \approx 1.128\ \text{V}$$
$$V_{E3} \approx 0.428\ \text{V}$$
$$I_{E3} \approx 0.522\ \text{mA}$$
$$V_{C3} \approx 8.77\ \text{V}$$

$$V_{CE3} \approx 8.34 \text{ V}$$

2.4 Bias of Q_4

The output stage is biased by $R_5 = 3$ MΩ and $R_{12} = 100$ kΩ.

$$V_{TH4} = 15 \cdot \frac{100 \text{ k}\Omega}{3 \text{ M}\Omega + 100 \text{ k}\Omega} = 0.484 \text{ V}$$
$$R_{TH4} = 3 \text{ M}\Omega \parallel 100 \text{ k}\Omega = 96.8 \text{ k}\Omega$$

For the output emitter follower,

$$V_{E4} = V_{B4} - 0.7$$
$$I_{E4} = \frac{V_{E4} + 15}{1 \text{ k}\Omega} = \frac{V_{B4} + 14.3}{1 \text{ k}\Omega}$$
$$I_{B4} = \frac{I_{E4}}{201}$$
$$V_{B4} = V_{TH4} - I_{B4} R_{TH4}$$

Solving gives

$$V_{B4} \approx -4.32 \text{ V}$$
$$V_{E4} \approx -5.02 \text{ V}$$
$$I_{E4} \approx \frac{-5.02 + 15}{1 \text{ k}\Omega} \approx 9.98 \text{ mA}$$

This relatively large current is intentional. The load current required for a 2.5 Vpp output across 200 Ω is

$$I_{L,peak} = \frac{1.25 \text{ V}}{200 \text{ Ω}} = 6.25 \text{ mA}$$

Therefore, biasing Q_4 near 10 mA gives the output stage enough current capability to drive the load linearly.

Small-signal analysis

Using

$$g_m = \frac{I_C}{V_T}, r_e = \frac{1}{g_m}, r_\pi = \frac{\beta}{g_m}$$

with $V_T \approx 25$ mV, the small-signal parameters become:

For Q_1, with $I_{C1} \approx 0.225$ mA,

$$g_{m1} \approx \frac{0.225 \text{ mA}}{25 \text{ mV}} = 9.0 \text{ mS}$$

$$r_{e1} \approx 111\ \Omega$$
$$r_{\pi 1} \approx 22.3\ \text{k}\Omega$$

For Q_2 and Q_3, with $I_C \approx 0.52$ mA,

$$g_{m2} \approx g_{m3} \approx \frac{0.52\ \text{mA}}{25\ \text{mV}} = 20.8\ \text{mS}$$
$$r_{e2} \approx r_{e3} \approx 48\ \Omega$$
$$r_{\pi 2} \approx r_{\pi 3} \approx 9.6\ \text{k}\Omega$$

For Q_4, with $I_C \approx 9.98$ mA,

$$g_{m4} \approx \frac{9.98\ \text{mA}}{25\ \text{mV}} = 0.399\ \text{S}$$
$$r_{e4} \approx 2.5\ \Omega$$
$$r_{\pi 4} \approx 501\ \Omega$$

These numbers explain the overall behavior of the circuit. The input transistor has a moderate r_π, but because its emitter sees a large resistance multiplied by $(\beta+1)$, the input resistance becomes very large. The gain stages have enough transconductance to produce voltage gain, and the output transistor has an extremely small r_e, which is exactly what is needed for low output impedance.

Input impedance

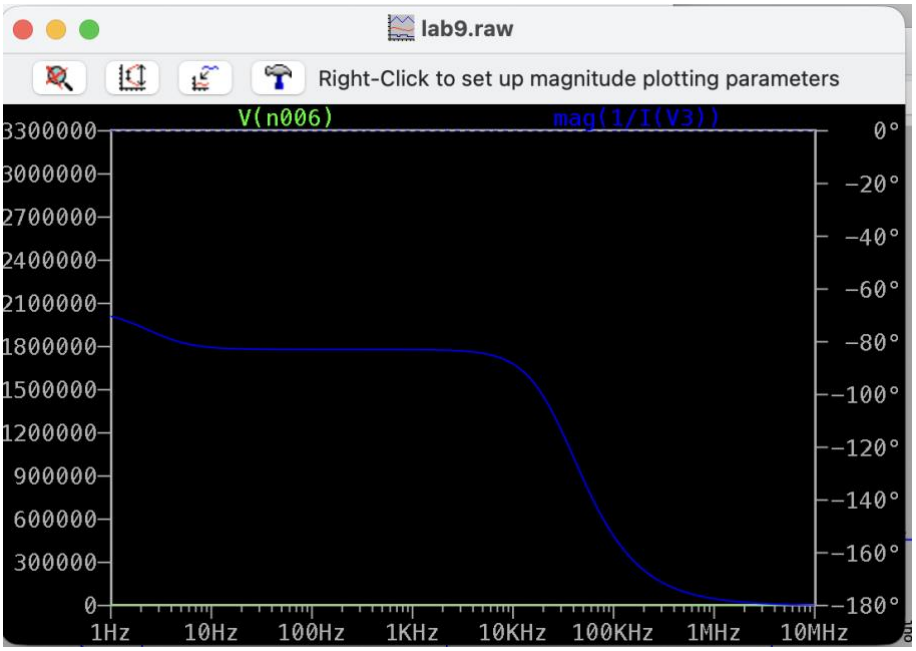


Figure 2. Input Impedance

The input resistance seen at the base of Q_2 is

$$R_{in,b2} \approx r_{\pi 2} + (\beta + 1)R_9$$
$$R_{in,b2} \approx 9.6\ \text{k}\Omega + 201(820\ \Omega) \approx 174\ \text{k}\Omega$$

Including the bias network,

$$R_{in,Q2} = R_{in,b2} \parallel (R_{15} \parallel R_8)$$
$$R_{in,Q2} \approx 174\ \text{k}\Omega \parallel 90.9\ \text{k}\Omega \approx 59.8\ \text{k}\Omega$$

The emitter of Q_1 therefore sees

$$R_{E,eff1} = R_7 \parallel R_{in,Q2}$$
$$R_{E,eff1} = 51\ \text{k}\Omega \parallel 59.8\ \text{k}\Omega \approx 27.5\ \text{k}\Omega$$

So the resistance looking into the base of Q_1 is

$$R_{in,b1} \approx r_{\pi 1} + (\beta + 1)R_{E,eff1}$$
$$R_{in,b1} \approx 22.3\ \text{k}\Omega + 201(27.5\ \text{k}\Omega) \approx 5.55\ \text{M}\Omega$$

Finally, including the input divider,

$$Z_{in} \approx R_{in,b1} \parallel (R_1 \parallel R_6)$$
$$Z_{in} \approx 5.55\ \text{M}\Omega \parallel 2.5\ \text{M}\Omega \approx 1.72\ \text{M}\Omega$$

This matches the LTspice result well, where the midband input impedance is about 1.75 to 1.8 MΩ. Therefore, the design satisfies the specification $Z_{in} \geq 1$ MΩ.

Midband gain

The first stage is an emitter follower, so its gain is close to unity:

$$A_{v1} \approx \frac{R_{E,eff1}}{R_{E,eff1} + r_{e1}} = \frac{27.5\ \text{k}\Omega}{27.5\ \text{k}\Omega + 111\ \Omega} \approx 0.996$$

For Q_2, the collector is loaded by the input of the next stage. The effective collector resistance is approximately

$$R_{C,eff2} = R_2 \parallel R_{in,Q3}$$

Since Q_3 is similar to Q_2, $R_{in,Q3} \approx 59.8$ kΩ, so

$$R_{C,eff2} = 12\ \text{k}\Omega \parallel 59.8\ \text{k}\Omega \approx 10.0\ \text{k}\Omega$$

Therefore,

$$A_{v2} \approx -\frac{R_{C,eff2}}{R_9 + r_{e2}} \approx -\frac{10.0\ \text{k}\Omega}{820\ \Omega + 48\ \Omega} \approx -11.5$$

The third stage is similar, but it is loaded by the input of Q_4. Because the output stage is heavier, the effective input resistance of Q_4 is smaller, and the gain of Q_3 becomes slightly lower. The result is approximately

$$A_{v3} \approx -9.36$$

The output stage is again an emitter follower. With

$$R_{E,eff4} = R_{13} \parallel R_L = 1\ \text{k}\Omega \parallel 200\ \Omega = 166.7\ \Omega$$

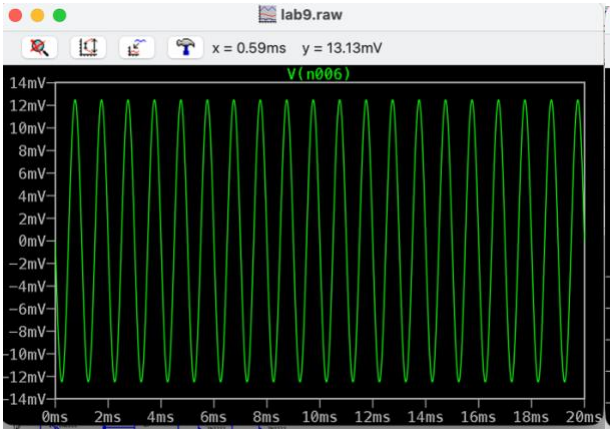
its voltage gain is

$$A_{v4} \approx \frac{R_{E,eff4}}{R_{E,eff4} + r_{e4}} = \frac{166.7}{166.7 + 2.5} \approx 0.985$$

Thus the total midband gain is

$$A_v \approx A_{v1}A_{v2}A_{v3}A_{v4}$$
$$A_v \approx (0.996)(-11.5)(-9.36)(0.985) \approx 106$$

This agrees very well with the AC simulation, where the midband gain is around 106 to 107 V/V. Since the acceptable range is 85 V/V to 115 V/V, the gain requirement is satisfied.



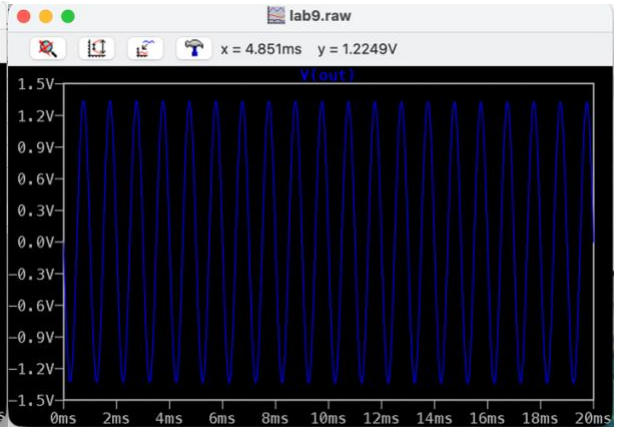


Figure 3 and 4. Input and output waves.

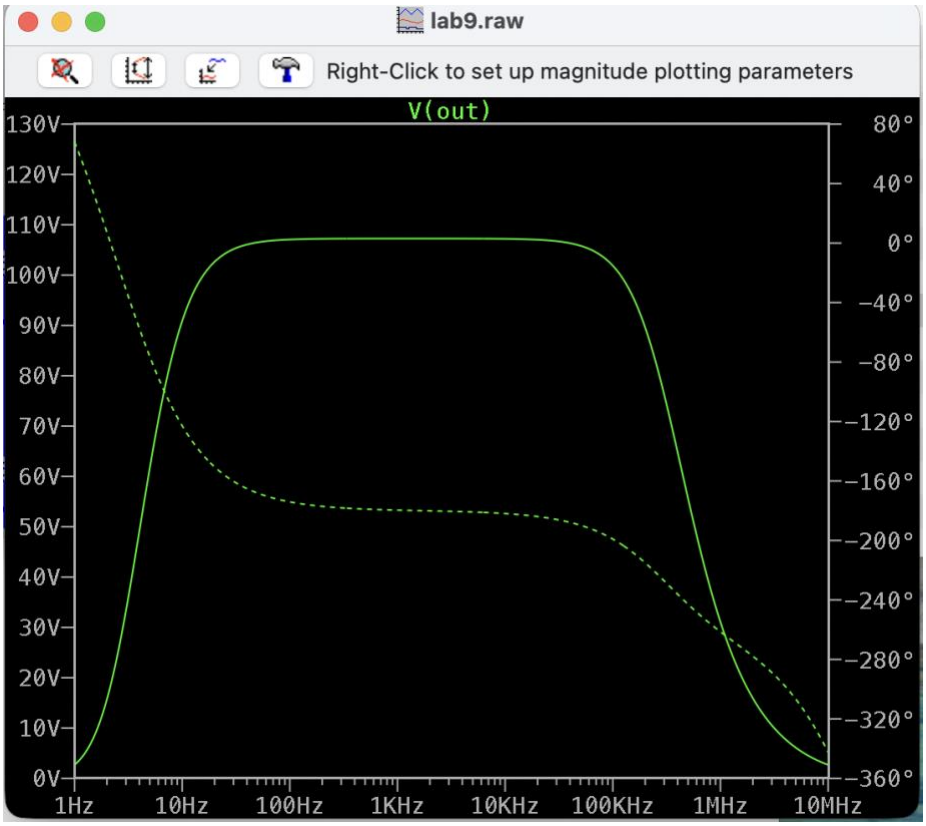


Figure 5. Midband Gain.

Output impedance

The output impedance is dominated by the emitter follower Q_4. A useful approximation is

$$Z_{out} \approx \left(r_{e4} + \frac{R_{B,source}}{\beta + 1}\right) \parallel R_{13}$$

Using the base-side driving resistance seen by Q_4, the result is approximately

$$Z_{out} \approx 50\ \Omega$$

The LTspice simulation gives a midband output impedance of approximately $47\ \Omega$, which is consistent with the hand calculation. Therefore, the circuit clearly satisfies the specification $Z_{out} \leq 100\ \Omega$.

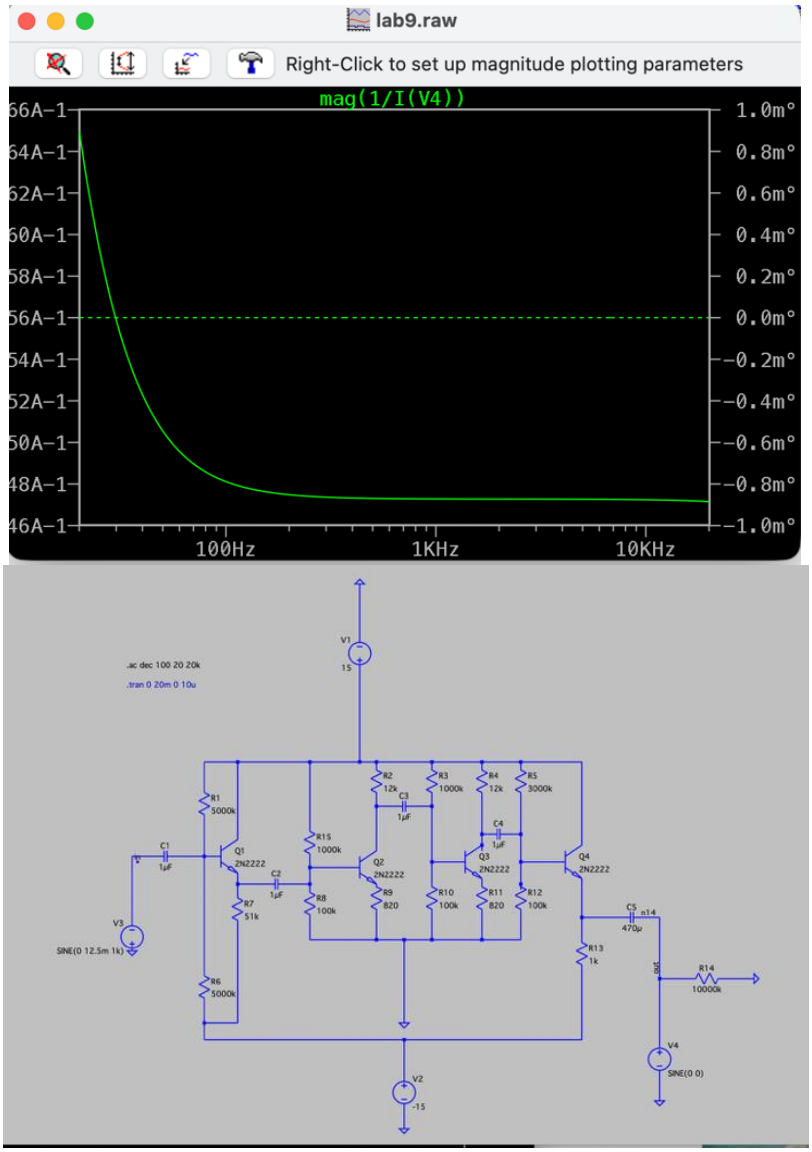


Figure 6 and 7. Output Impedance and Test Voltage For Determining Zout.

Low-frequency response

The lower cutoff frequency is mainly determined by the coupling capacitors.

For the input capacitor C_1, $f_{C1} \approx \frac{1}{2\pi C_1 Z_{in}} = \frac{1}{2\pi(1\ \mu\text{F})(1.72\ \text{M}\Omega)} \approx 0.09$ Hz

For C_2, $f_{C2} \approx \frac{1}{2\pi C_2(R_{left}+R_{right})} \approx \frac{1}{2\pi(1\ \mu\text{F})(69.8\ \text{k}\Omega)} \approx 2.28$ Hz

For C_3, $f_{C3} \approx \frac{1}{2\pi(1\ \mu\text{F})(71.8\ \text{k}\Omega)} \approx 2.22$ Hz

For C_4, $f_{C4} \approx \frac{1}{2\pi(1\ \mu\text{F})(37.2\ \text{k}\Omega)} \approx 4.28$ Hz

For the output capacitor C_5, $f_{C5} \approx \frac{1}{2\pi C_5(Z_{out}+R_L)}$. $f_{C5} \approx \frac{1}{2\pi(470\ \mu\text{F})(47+200)} \approx 1.37$ Hz

Because several low-frequency poles are present, the overall lower 3-dB frequency is somewhat higher than any single pole alone. From the Bode plot, the overall lower cutoff is still only in the single-digit-hertz range, clearly below the required 20 Hz.

High-frequency response and effect of small-signal parasitics

At high frequency, the main limitation comes from transistor parasitic capacitances, especially the base-collector capacitance of the common-emitter stages. In Q_2 and Q_3, this capacitance is multiplied by the Miller effect:

$$C_M \approx C_\mu(1+|\ A_v\ |)$$

Since both common-emitter stages have gains around 10, the effective input capacitance of these stages becomes much larger than the physical transistor capacitance. This creates high-frequency poles and eventually reduces the overall gain. However, because the total gain is distributed over two moderate-gain stages rather than one extremely high-gain stage, the high-frequency response remains good. Also, the emitter followers at the input and output help isolate the high-impedance nodes and reduce severe loading. The AC simulation shows that the upper 3-dB frequency is far above 20 kHz, on the order of several hundred kilohertz, so the bandwidth requirement is easily satisfied.

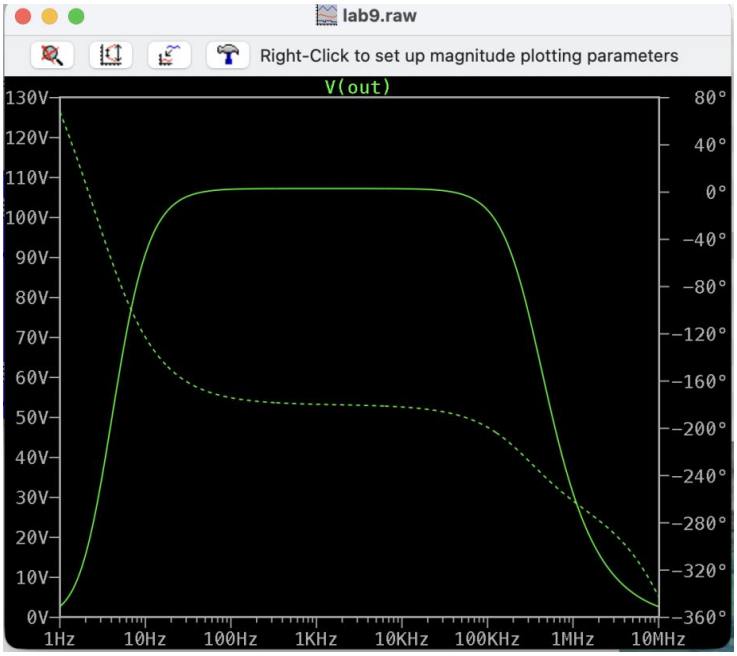


Figure 8. Frequency Response and 3 db Points.

Function of each transistor

Q_1 is the input buffer. It takes the incoming signal through C_1, presents a high input resistance to the source, and drives the next stage through C_2. Its voltage gain is close to 1, but its current gain is high. Its main purpose is not voltage amplification but isolation and impedance buffering.

Q_2 is the first voltage-gain stage. It is a common-emitter amplifier, so small changes in base-emitter voltage cause amplified changes in collector voltage. The collector resistor R_2 converts collector current changes into a voltage swing. The emitter resistor R_9 introduces degeneration, which stabilizes the operating point and improves linearity.

Q_3 is the second voltage-gain stage. It works in the same way as Q_2. This stage provides additional gain, and because it introduces a second inversion, the total amplifier becomes non-inverting.

Q_4 is the output buffer. It is an emitter follower biased at significantly higher current than the previous stages. This makes its small-signal emitter resistance very small, which is why it can drive the $200\ \Omega$ load while maintaining low output impedance and low distortion.

LTspice verification

The transient simulation uses a 1 kHz sinusoidal input with amplitude 12.5 mV. The output waveform is approximately 2.7 Vpp, which corresponds to about 1.35 V peak. Therefore, the transient gain is approximately

$$A_v \approx \frac{1.35\ \text{V}}{12.5\ \text{mV}} = 108$$

This confirms that the design meets the required output swing without visible nonlinear distortion.

The AC transfer plot shows a midband gain of about $106–107$ V/V, a lower cutoff below 20 Hz, and an upper cutoff far above 20 kHz. The input impedance plot gives a midband value around $1.75–1.8$ MΩ, and the output impedance plot gives a midband value around 47 Ω. All of these values satisfy the original design criteria.

Conclusion

A four-stage BJT amplifier using 2N2222 transistors was successfully designed to meet all of the required specifications. The first emitter follower provides high input impedance, the two common-emitter stages provide the required voltage gain while preserving an overall non-inverting response, and the final emitter follower provides low output impedance and sufficient current drive for the 200 Ω load. The resistor and capacitor values were selected carefully based on gain, biasing, bandwidth, impedance, and load-driving requirements. Hand calculations predict a total gain close to 106 V/V, an input impedance of about 1.72 MΩ, and an output impedance of about 50 Ω. These results are consistent with LTspice, which shows approximately 106–107 V/V midband gain, about 1.8 MΩ input impedance, about 47 Ω output impedance, and a clean 2.5 Vpp output at 1 kHz. Therefore, the design satisfies the gain, bandwidth, impedance, and output swing specifications.

3. Data/Results

3.1 Part 1: Input Stage / Common-Collector Buffer

Building Block

The first checkpoint of the in-lab assignment was to build and test the input stage of the multistage amplifier. The lab handout specifically suggests implementing and testing the amplifier stage by stage instead of building the complete circuit at once. Therefore, the first stage was constructed separately and tested before connecting it to the amplification stages. The purpose of this stage was to satisfy the high input impedance requirement of the overall amplifier and to prevent the signal source from being loaded by the following stages. The full amplifier was required to have an input impedance greater than 1 MΩ, so the input stage had to be designed carefully.

For this purpose, Q1 was connected in common-collector configuration, also known as an emitter follower. A common-collector stage has a voltage gain close to unity, high input impedance, and low output impedance compared with the source resistance. Because of these properties, it is commonly used as a buffer stage rather than as a voltage-gain stage. In this circuit, Q1 does not significantly amplify the voltage. Instead, it isolates the signal source from the rest of the amplifier and allows the following common-emitter stages to be driven without reducing the input impedance of the whole circuit. This is why the pre-lab design also used Q1 as the input buffer stage.

Components used in this stage:

Q1 = 2N2222 BJT
R1 = 5 MΩ
R6 = 5 MΩ
R7 = 51 kΩ
C1 = 1 µF
C2 = 1 µF
Rtest = 1 MΩ, used only for input impedance measurement
Representative load = 27 kΩ, used only while testing Q1 separately

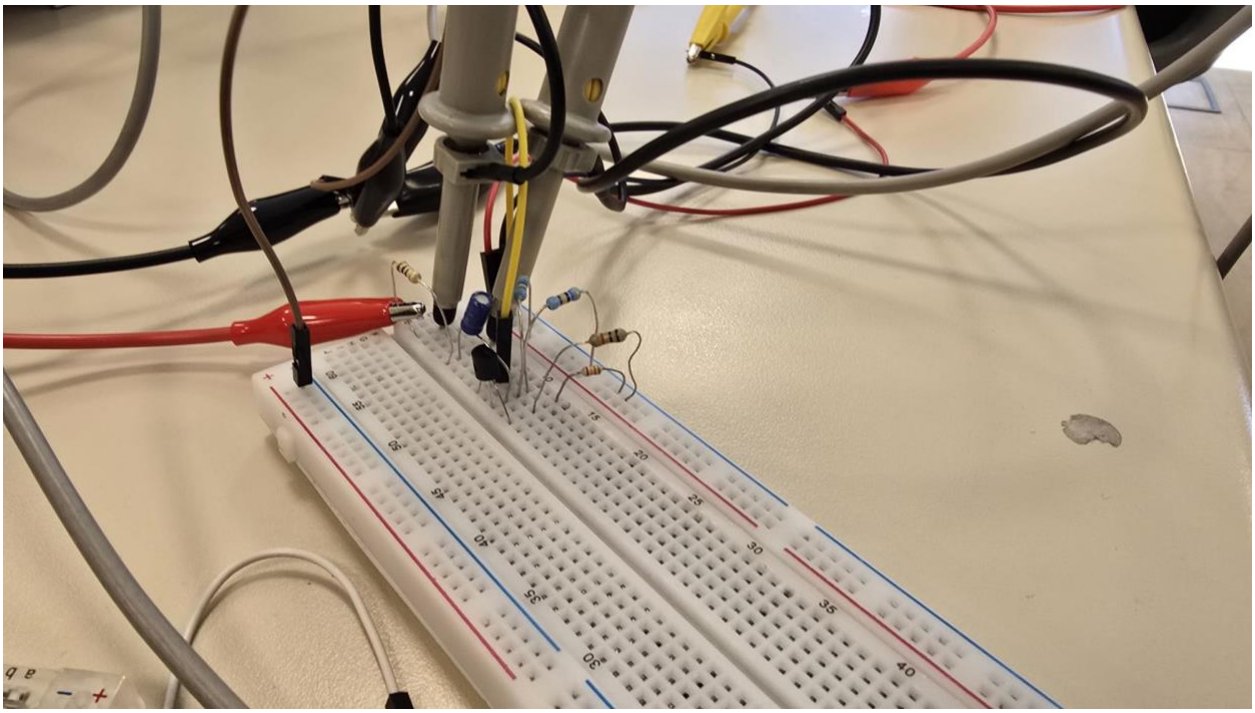


Figure 9. Breadboard Implementation

The base of Q1 was biased using R1 and R6. These two resistors were connected between +15 V and −15 V and formed a voltage divider. Since R1 and R6 were both selected as 5 MΩ, the ideal unloaded base voltage was approximately 0 V. The main reason for choosing very large resistor values was to keep the input impedance high. The parallel combination of the two bias resistors is

R1 || R6 = 5 MΩ || 5 MΩ = 2.5 MΩ

This value is already higher than the required 1 MΩ input impedance. If smaller bias resistors had been used, the source would have seen a lower resistance at the input node, and the input impedance of the amplifier would have decreased. Therefore, R1 and R6 were selected mainly according to the input impedance requirement.

The emitter resistor R7 was selected as 51 kΩ. In a common-collector stage, the emitter resistor helps set the DC emitter current and also affects the input resistance seen from the base. The resistance at the emitter is reflected to the base by approximately a factor of β + 1. Therefore, using a relatively large emitter resistor helps increase the input resistance of the stage. At the same time, R7 keeps the current of Q1 relatively low, which is acceptable because the first stage is only used as a buffer and does not need to drive the 200 Ω load directly. In the pre-lab calculation, the emitter current of Q1 was estimated to be approximately 0.225 mA, which is sufficient for input buffering.

C1 was used as the input coupling capacitor. Its purpose is to pass the AC input signal to the base of Q1 while blocking DC current between the function generator and the biasing network. This is necessary because the lab specification states that there should be no DC current flow through the signal source. C2 was used as the output coupling capacitor of the input stage. It passes the AC signal from the emitter of Q1 to the next stage while blocking the DC bias voltage of the input stage from affecting the DC operating point of the following common-emitter stage. Therefore, C1 and C2 are both coupling capacitors used for AC signal transfer and DC isolation. In the pre-lab design, both capacitors were selected as 1 µF because the resistances around these capacitors are relatively large, so the resulting low-frequency poles remain well below the required lower cutoff frequency of 20 Hz.

Measurements

During the lab measurement, Q1 was tested separately from the rest of the amplifier. Since the next amplification stage was not connected yet, a 27 kΩ resistor was added at the output as a representative load. This load was used to approximate the loading effect that the next stage would create in the complete multistage amplifier. Therefore, the measurement did not test Q1 under a completely unloaded condition. Instead, it tested the common-collector buffer in a more realistic condition, where the emitter side is loaded by an approximate next-stage resistance.

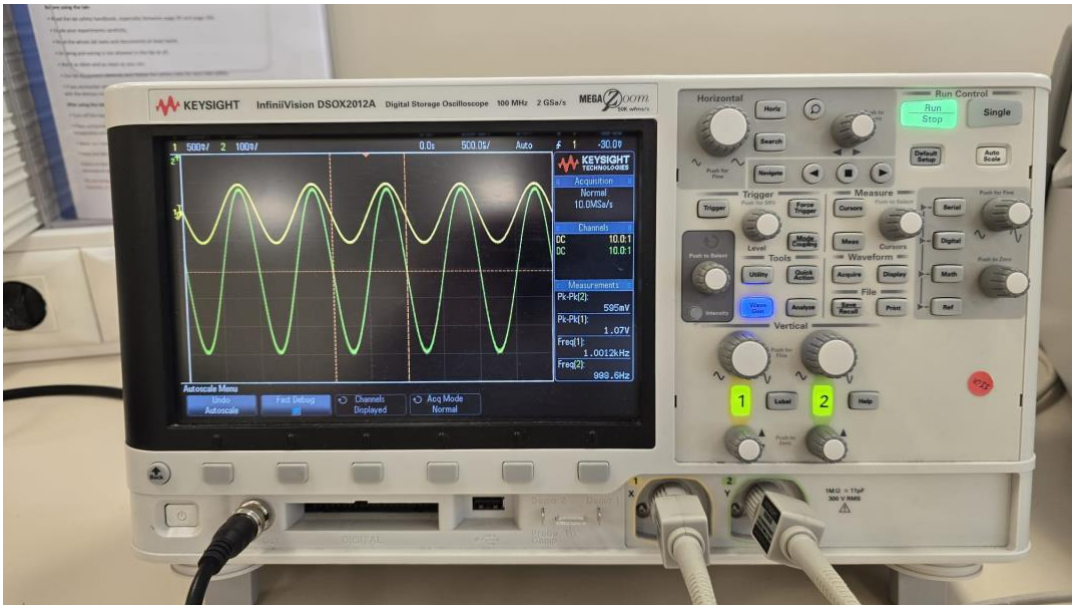


Figure 10. Oscilloscope measurement used for the input impedance test of the input stage.

To measure the input impedance of this stage, a 1 MΩ test resistor was connected in series with the input of the circuit. A 1 Vpp sinusoidal test signal at approximately 1 kHz was applied across the series combination of Rtest and the amplifier input impedance. The oscilloscope was used to measure the voltage division between the known 1 MΩ resistor and the unknown input impedance. The measured voltage across the input impedance was approximately 595 mVpp. Since the total applied voltage across Rtest and Zin was approximately 1 Vpp, the voltage across the 1 MΩ test resistor was VRtest = 1 Vpp − 0.595 Vpp = 0.405 Vpp

Using the voltage divider relationship, VZin / Vin = Zin / (Rtest + Zin) the input impedance can be found as Zin = Rtest × VZin / Vrtest Substituting the measured values, Zin = 1 MΩ × 0.595 / 0.405
Zin ≈ 1.47 MΩ

Therefore, the measured input impedance of the first stage was approximately 1.47 MΩ. This value is above the required 1 MΩ limit, so the input stage satisfies the input impedance specification. The result also shows that the common-collector stage can still provide high input impedance even when it is loaded by a representative 27 kΩ load.

The oscilloscope measurement showed that the input and output waveforms were both sinusoidal and had approximately the same frequency, close to 1 kHz. Since this stage is an emitter follower, a large voltage gain was not expected. The goal was not to amplify the voltage but to preserve the signal shape and provide buffering. The measured result was consistent with this expectation because the stage passed the signal without visible nonlinear distortion while maintaining a high input impedance.

The measured result was then compared with the pre-lab calculation and LTspice simulation. In the pre-lab, the expected input impedance of the overall input stage was calculated as approximately 1.72 MΩ. The LTspice result was also close to this value, around 1.75–1.8 MΩ. In the lab, the measured value was approximately 1.47 MΩ. Although this is lower than the pre-lab and simulation results, it still satisfies the design requirement.

There are several reasons why the measured result did not exactly match the theoretical and LTspice values. First, the theoretical calculation and LTspice model assumed a typical β value of approximately 200 for the 2N2222 transistor. However, the real 2N2222 transistor used in the lab can have a much lower β value, and β can vary significantly between different transistors. This is important because the input resistance of an emitter follower strongly depends on β. The resistance at the emitter is reflected to the base by approximately β + 1. Therefore, if the real transistor β is lower than the assumed value, the actual input impedance becomes lower than the theoretical value.

Second, Q1 was tested separately with a 27 kΩ representative load at its output. In the complete pre-lab calculation, the loading of the next stage was estimated theoretically. In the lab, however, the 27 kΩ resistor created a real load on the emitter of Q1. Since the effective emitter resistance directly affects the input impedance of a common-collector stage, adding this load reduces the emitter-side resistance and therefore also reduces the measured input impedance. This helps explain why the measured Zin was lower than the LTspice value.

Third, practical breadboard effects can also create differences between measurement and simulation. The real circuit includes contact resistance, wire resistance, parasitic capacitance, oscilloscope probe loading, and small leakage currents. These effects become more noticeable because the bias resistors R1 and R6 are very large, 5 MΩ each. Large resistor values are useful for high input impedance, but they also make the circuit more sensitive to noise, leakage, and measurement loading. These nonideal effects are not fully included in the ideal hand calculations.

As a result, the small difference between theory, simulation, and lab measurement was expected. The important point is that the measured input impedance was still higher than 1 MΩ and the waveform was not distorted. Therefore, the input stage successfully performed its intended function as a high-input-impedance common-collector buffer. This confirms that the first building block of the multistage amplifier worked properly and was suitable for connection to the following amplification stages.

3.2 Part 2: Amplification Stage

Building Block

The second part of the in-lab work was the amplification stage of the multistage BJT amplifier. After the input buffer stage was tested separately, the main voltage-gain block was implemented on the breadboard and measured. The purpose of this block was to provide most of the required voltage gain while keeping the output waveform sinusoidal and avoiding clipping. According to the lab specifications, the complete amplifier was expected to have a non-inverting midband voltage gain of 100 ± 15%, a lower 3-dB frequency below 20 Hz, and an upper 3-dB frequency above 20 kHz. The lab instructions also recommend building and testing the circuit stage by stage, so this amplification stage was verified before connecting the final output buffer stage.

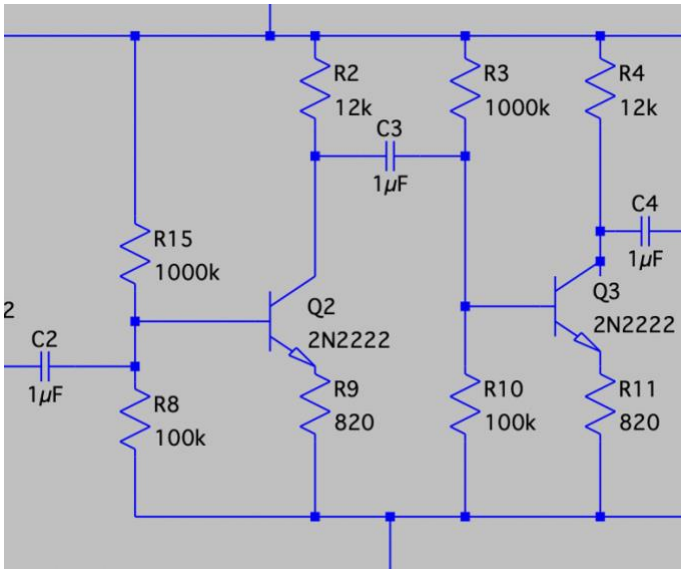
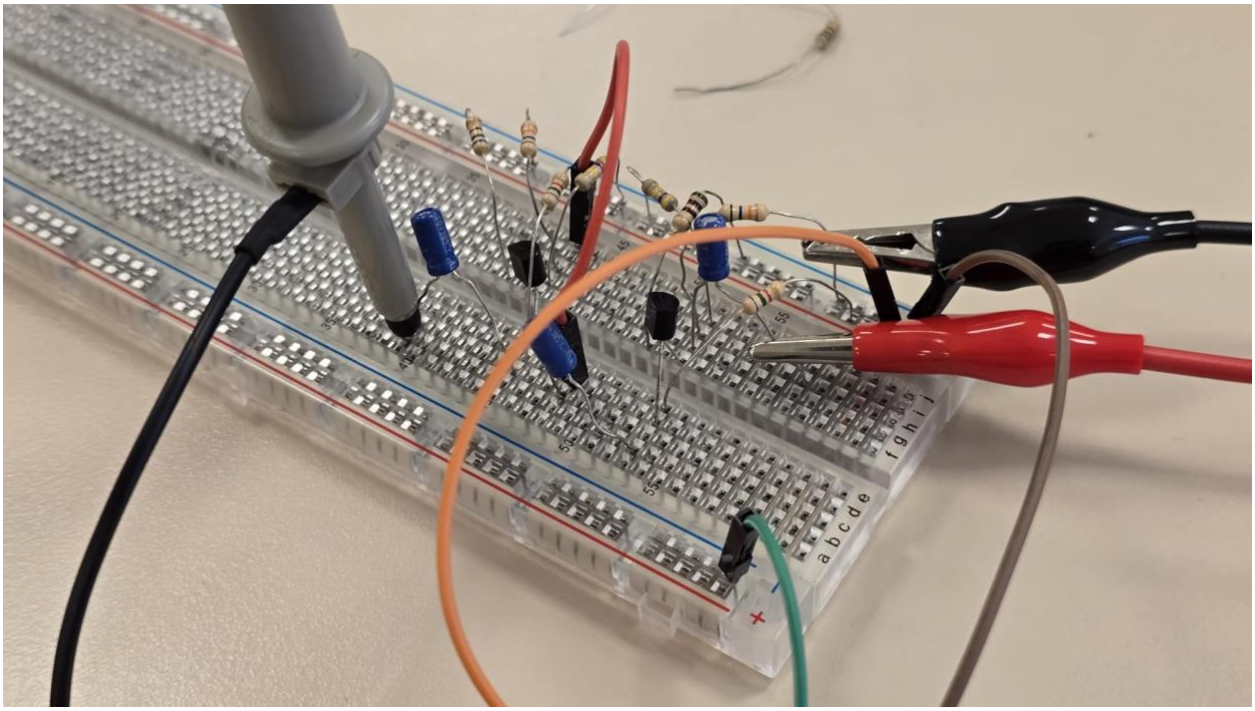


Figure 11. LTspice schematic of the multistage BJT amplifier used as the reference design.


Figure 12. Breadboard implementation of the amplification stage during in-lab testing.

The amplification stage mainly consists of Q2 and Q3, which are both 2N2222 BJTs connected in common-emitter configuration. In the complete amplifier, Q1 is the input buffer, Q2 and Q3 are the voltage-gain stages, and Q4 is the output buffer. Therefore, Q2 and Q3 are responsible for increasing the signal amplitude. A common-emitter amplifier was selected for these two stages because it provides significant voltage gain. However, instead of trying to obtain the full gain of about 100 V/V from one transistor, the gain was divided between two common-emitter stages. This makes the design more stable, less sensitive to transistor parameter variation, and better for bandwidth.

Each common-emitter stage introduces a 180° phase shift. Since the amplification block contains two common-emitter stages, the signal is inverted twice. As a result, the final output of the amplification block is non-inverting with respect to the input signal. This is important because the required overall amplifier response is non-inverting. In the pre-lab design, Q2 and Q3 were also described as the main voltage-gain stages that provide most of the required amplification.
The components used in this stage were Q2 = 2N2222, Q3 = 2N2222, R15 = 1 MΩ, R8 = 100 kΩ, R2 = 12 kΩ, R9 = 820 Ω, R3 = 1 MΩ, R10 = 100 kΩ, R4 = 12 kΩ, R11 = 820 Ω, C2 = 1 µF, C3 = 1 µF, and C4 = 1 µF. Q2 forms the first common-emitter gain stage, where R15 and R8 bias the base, R2 is the collector resistor, and R9 is the emitter resistor. Q3 forms the second common-emitter gain stage, where R3 and R10 bias the base, R4 is the collector resistor, and R11 is the emitter resistor. C2 couples the signal from the input buffer to Q2, C3 couples the output of Q2 to the input of Q3, and C4 couples the output of Q3 to the next stage.

The bias dividers R15–R8 and R3–R10 were selected as 1 MΩ and 100 kΩ. These resistor pairs create base-bias voltages for Q2 and Q3 from the +15 V supply. Ideally, the unloaded base voltage is VB = 15 V × 100 kΩ / (1 MΩ + 100 kΩ) = 1.36 V. In the real circuit, the base current of the transistor loads this divider slightly, so the actual base voltage becomes lower than the ideal value. In the pre-lab calculation, the base voltages of Q2 and Q3 were estimated as approximately 1.13 V. Since a silicon BJT has

approximately VBE ≈ 0.7 V, the emitter voltage becomes VE ≈ 1.13 V − 0.7 V = 0.43 V. With R9 = R11 = 820 Ω, the emitter current becomes IE ≈ 0.43 V / 820 Ω = 0.52 mA. This current level gives enough transconductance for voltage gain while keeping the transistors safely in the active region.

The collector resistors R2 and R4 were selected as 12 kΩ. In a common-emitter amplifier, the collector resistor converts collector current changes into output voltage changes. A larger collector resistor would increase the voltage gain, but it would also create a larger DC voltage drop and reduce the available output swing. This would make clipping more likely. A smaller collector resistor would give more voltage headroom but would reduce the gain. Therefore, 12 kΩ was chosen as a balance between obtaining enough gain and keeping enough collector voltage swing.

The emitter resistors R9 and R11 were selected as 820 Ω. These resistors provide emitter degeneration. Emitter degeneration reduces gain compared with a bypassed emitter resistor, but it improves bias stability, linearity, and predictability. This was important in the lab because the real 2N2222 transistors do not necessarily have the same β value as the LTspice model. In the pre-lab hand calculations, β was assumed to be approximately 200, but in the actual lab the β value can be much lower. Because of the emitter resistors, the gain depends more on external resistor values and less directly on β. Therefore, R9 and R11 make the amplifier more reliable in practice. The value 820 Ω was chosen instead of a larger value such as 1 kΩ because the gain still needed to stay close to 100 V/V. A larger emitter resistor would reduce the gain too much, while a much smaller emitter resistor would make the stage more nonlinear and more sensitive to transistor variation.

The approximate voltage gain of a common-emitter stage with an unbypassed emitter resistor is $A_v \approx -R_{C,eff} / (R_E + r_e)$, where $R_{C,eff}$ is the effective collector resistance, R_E is the external emitter resistor, and r_e is the small-signal emitter resistance. The negative sign means that one common-emitter stage inverts the signal. From the pre-lab bias point, IC ≈ 0.52 mA, so gm ≈ IC / VT = 0.52 mA / 25 mV = 20.8 mS and re ≈ 1 / gm = 1 / 20.8 mS ≈ 48 Ω. For Q2, the collector is loaded by the input resistance of Q3, so the effective collector resistance is not simply 12 kΩ. In the pre-lab, $R_{C,eff2}$ was estimated as about 10 kΩ, so Av2 ≈ −10 kΩ / (820 Ω + 48 Ω) = −11.5. The second common-emitter stage has a similar structure but is loaded by the following stage, so its gain was estimated as Av3 ≈ −9.36. Therefore, the gain of the two common-emitter stages is approximately Av2 × Av3 = (−11.5)(−9.36) = +107.6. The positive sign shows that the two inversions cancel each other. This theoretical result is close to the target gain and agrees with the pre-lab expectation that the amplifier should have a midband gain around 106–107 V/V.

The coupling capacitors C2, C3, and C4 were selected as 1 µF. Their purpose is to pass the AC signal from one stage to the next while blocking DC. This is necessary because each transistor stage has its own DC bias point. If the stages were directly connected without coupling capacitors, the collector DC voltage of one stage would disturb the base bias of the next stage and could push the transistor out of the active region. These capacitors also determine the low-frequency response. At low frequencies, the capacitor reactance increases, so less signal is transferred between stages and the gain decreases. In the pre-lab, the low-frequency poles associated with C2, C3, and C4 were estimated to be below 20 Hz, so 1 µF was expected to be large enough for the required lower cutoff specification.

At high frequencies, the gain is mainly limited by transistor parasitic capacitances and the Miller effect. In common-emitter stages, the base-collector capacitance is effectively multiplied by the gain of the stage. This increases the effective input capacitance and creates high-frequency poles. As the frequency increases, these capacitances begin to reduce the gain. However, because the total gain was divided

between two moderate-gain common-emitter stages instead of one very high-gain stage, the upper cutoff frequency remained much higher than 20 kHz.

Measurements

The amplification stage was tested using the waveform generator and oscilloscope. The input signal was a sinusoidal waveform with 25 mVpp amplitude and 0 V DC offset. This small input amplitude was selected because the expected gain was close to 100 V/V. With this gain, a 25 mVpp input should produce an output around 2.5 Vpp. If a larger input signal had been applied, the output could have exceeded the linear swing range of the transistors and caused clipping. Therefore, 25 mVpp was appropriate for measuring gain and bandwidth without forcing the circuit into nonlinear operation.

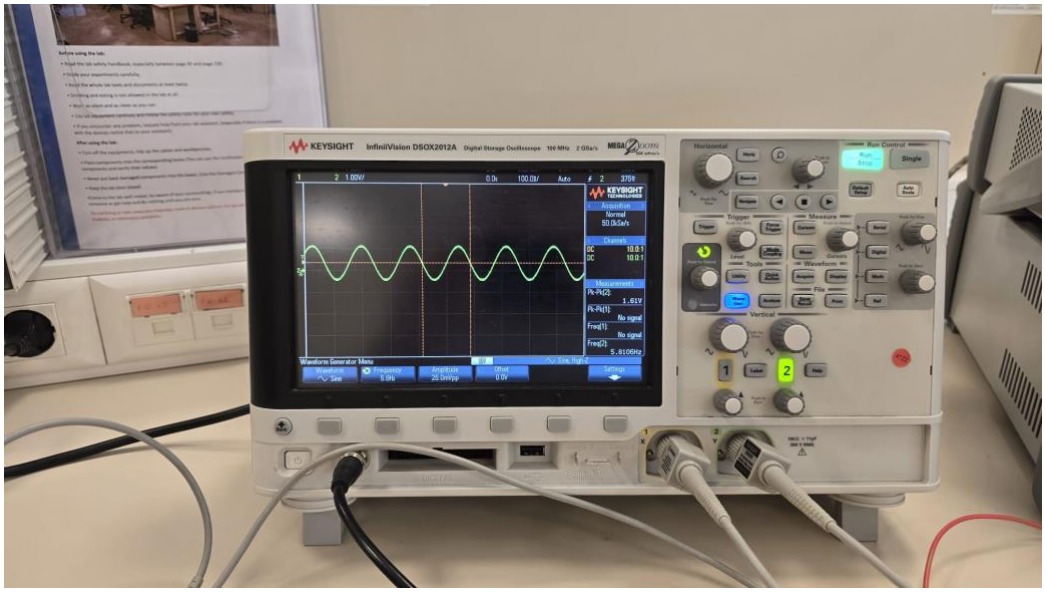


Figure 13.Oscilloscope measurement of the amplification stage near the lower 3-dB frequency.

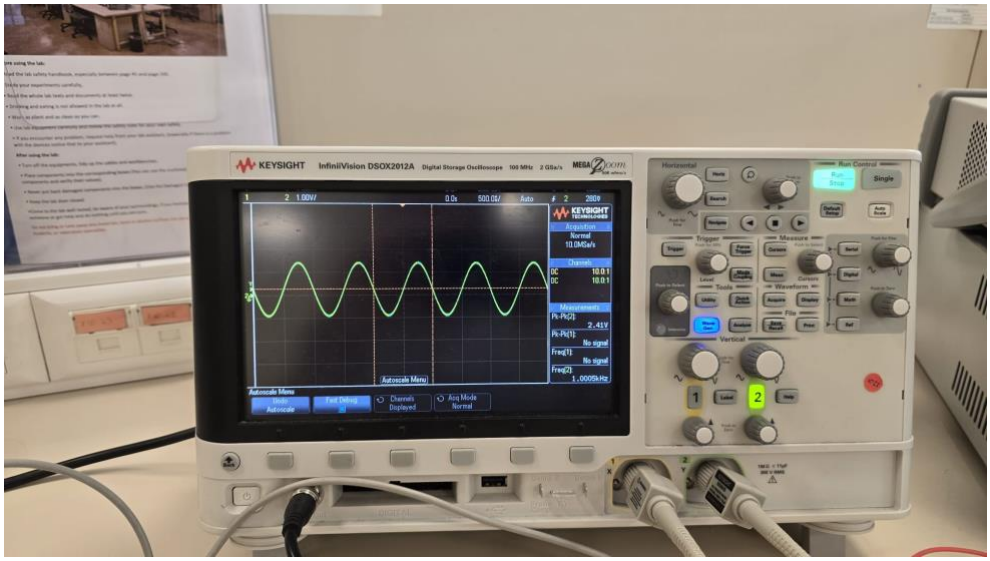


Figure 14. Oscilloscope measurement of the amplification stage at the 1 kHz midband frequency.

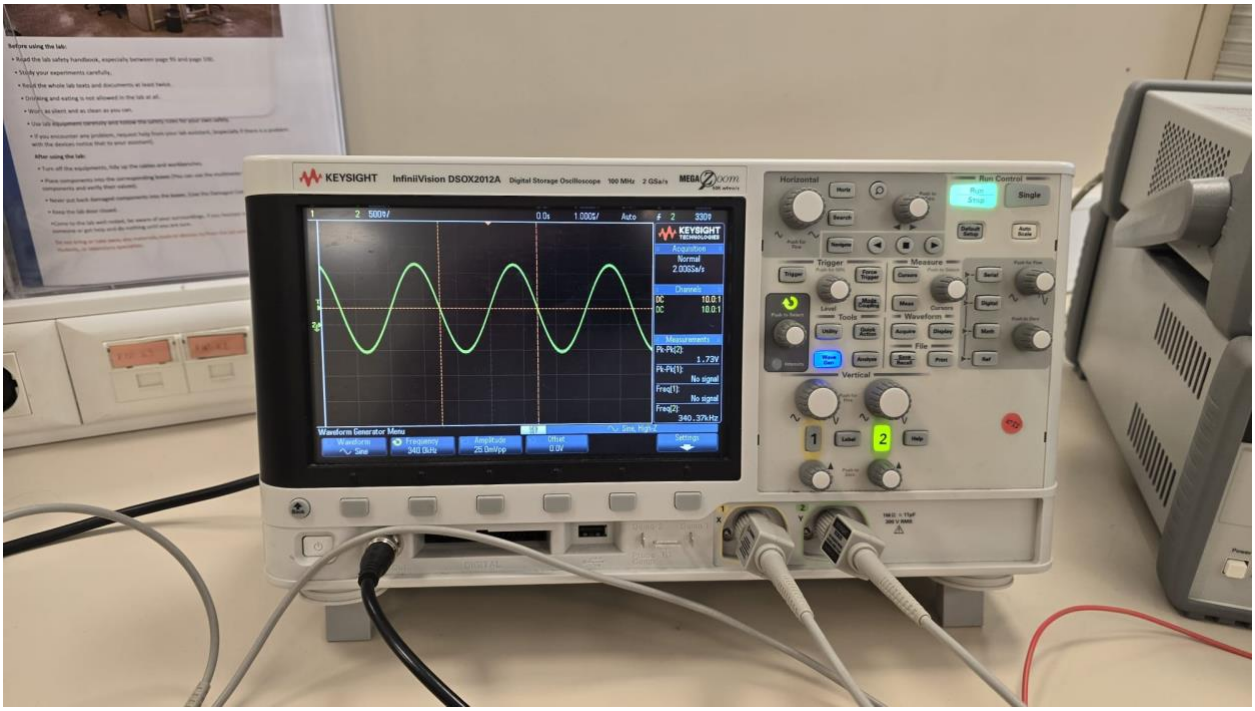


Figure 15. Oscilloscope measurement of the amplification stage near the upper 3-dB frequency.

At 1 kHz, the measured output voltage was 2.41 Vpp. Since 1 kHz is in the midband region, this value was used as the midband reference. The midband gain was Av = Vout / Vin = 2.41 Vpp / 25 mVpp = 2.41 / 0.025 = 96.4 V/V. This value is within the required gain range of 85 V/V to 115 V/V. The waveform was sinusoidal and did not show visible clipping, which indicates that Q2 and Q3 were biased properly and operating in the active region.

To find the 3-dB output level, the midband output voltage was divided by √2. Therefore, V3dB = Vmid / √2 = 2.41 Vpp / √2 = 1.70 Vpp. This means that the cutoff frequencies occur when the output amplitude drops from 2.41 Vpp to approximately 1.70 Vpp.

For the lower 3-dB measurement, the frequency was decreased until the output voltage became close to the calculated 3-dB level. At 5.81 Hz, the measured output voltage was 1.61 Vpp. The gain at this frequency was Av = Vout / Vin = 1.61 Vpp / 25 mVpp = 64.4 V/V. The expected 3-dB gain based on the midband gain is Av,3dB = 96.4 / √2 = 68.2 V/V. The measured value of 64.4 V/V is slightly lower than the ideal 3-dB gain, which means the exact lower 3-dB frequency is probably slightly higher than 5.81 Hz. However, this point is clearly in the lower cutoff region, and it proves that the lower 3-dB frequency is far below the required 20 Hz limit.

For the upper 3-dB measurement, the frequency was increased until the output voltage dropped close to the calculated 3-dB level. At approximately 340–380 kHz, the measured output voltage was about 1.73 Vpp. The gain at this frequency was Av = Vout / Vin = 1.73 Vpp / 25 mVpp = 69.2 V/V. This is very close to the calculated 3-dB gain of 68.2 V/V. Therefore, the upper 3-dB frequency of the amplification stage can be estimated to be in the range of approximately 340–380 kHz. This is much higher than the required upper cutoff specification of 20 kHz, so the amplification stage satisfies the high-frequency bandwidth requirement.

The measured values can be summarized as follows: at the lower cutoff region, f = 5.81 Hz, Vin = 25 mVpp, Vout = 1.61 Vpp, and Av = 64.4 V/V; at midband, f = 1.0005 kHz, Vin = 25 mVpp, Vout = 2.41 Vpp, and Av = 96.4 V/V; at the upper cutoff region, f ≈ 340–380 kHz, Vin = 25 mVpp, Vout = 1.73 Vpp, and Av = 69.2 V/V. The calculated 3-dB output voltage was 1.70 Vpp, so both the lower-frequency and upper-frequency measurements are close to the expected cutoff level.

Comparison with Theory and Pre-Lab Results

The measured midband gain was 96.4 V/V, while the pre-lab and LTspice result was around 106–107 V/V. The measured value is slightly lower, but it still satisfies the allowed gain range of 100 ± 15%. This difference is expected because the real circuit was built on a breadboard using real 2N2222 transistors, whose parameters can differ from the LTspice model.

The most important reason for the difference is β variation. In the pre-lab calculations, β was assumed to be around 200. In the lab, however, the actual β of the 2N2222 transistors can be much lower. A lower β causes more base current, which loads the bias divider more strongly. This can slightly change the base voltage, emitter current, collector current, and small-signal parameters. Since the voltage gain depends on the operating point and loading between stages, the real gain can be different from the theoretical value.

Emitter degeneration reduces this sensitivity but does not remove it completely. Because R9 and R11 are unbypassed, the gain is mainly controlled by the ratio $R_{C,eff} / (R_E + r_e)$. This makes the gain more predictable than a fully bypassed common-emitter stage. However, β still affects the input resistance of each stage and the loading between Q2 and Q3, so some difference between pre-lab and measurement is normal.

Component tolerances also affect the result. The gain depends strongly on R2, R4, R9, and R11. If R2 or R4 is slightly lower than 12 kΩ, the gain decreases. If R9 or R11 is slightly higher than 820 Ω, the gain also decreases. Since two common-emitter stages are cascaded, small errors in each stage multiply and can noticeably change the total gain.

The lower 3-dB result is mainly determined by the coupling capacitors C2, C3, and C4 and the resistances around them. At very low frequencies, the capacitive reactance of the coupling capacitors becomes larger, so the signal transfer between stages decreases. This explains why the output dropped to 1.61 Vpp at around 5.81 Hz. Since this frequency is below 20 Hz, the low-frequency design satisfies the lab requirement.
The upper 3-dB result is mainly caused by transistor parasitic capacitances, breadboard parasitic capacitance, oscilloscope probe capacitance, and the Miller effect in the common-emitter stages. At midband, these capacitances have little effect, but at high frequency they start to shunt signal current and reduce gain. This is why the output dropped to approximately 1.73 Vpp at around 340–380 kHz. Since this is much higher than 20 kHz, the high-frequency response is more than sufficient for the required specification.

In conclusion, the amplification stage successfully performed its function. At 1 kHz, the stage produced 2.41 Vpp output from a 25 mVpp input, giving a midband gain of 96.4 V/V. The lower cutoff region was measured near 5.81 Hz, which is below the required 20 Hz limit. The upper cutoff region was measured

around 340–380 kHz, which is far above the required 20 kHz limit. Therefore, the amplification stage satisfies both the gain and bandwidth requirements and is suitable to be connected to the output buffer stage.

3.3 Part 3: Output Stage

Building Block

The third part of the in-lab work was the output stage of the multistage BJT amplifier. After the input buffer and the amplification stage were tested, the final stage was implemented to drive the load with low output impedance. The overall amplifier specification required an output impedance lower than 100 Ω and a 2.5 Vpp output swing across a 200 Ω load without nonlinear distortion. Therefore, the purpose of this stage was not to provide additional voltage gain, but to act as a current buffer that can deliver enough current to the load while keeping the output voltage stable.

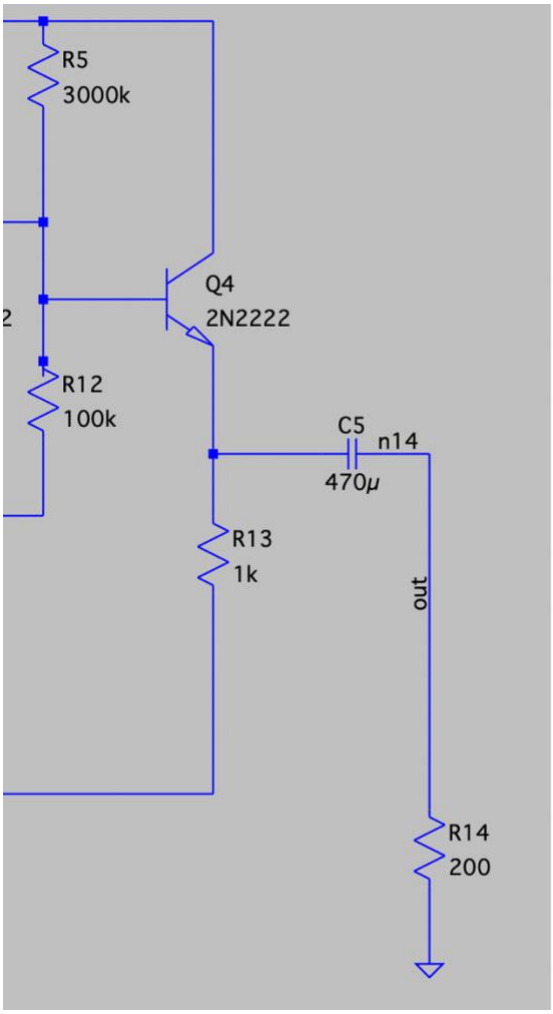


Figure 16. LTspice schematic of the common-collector output stage.

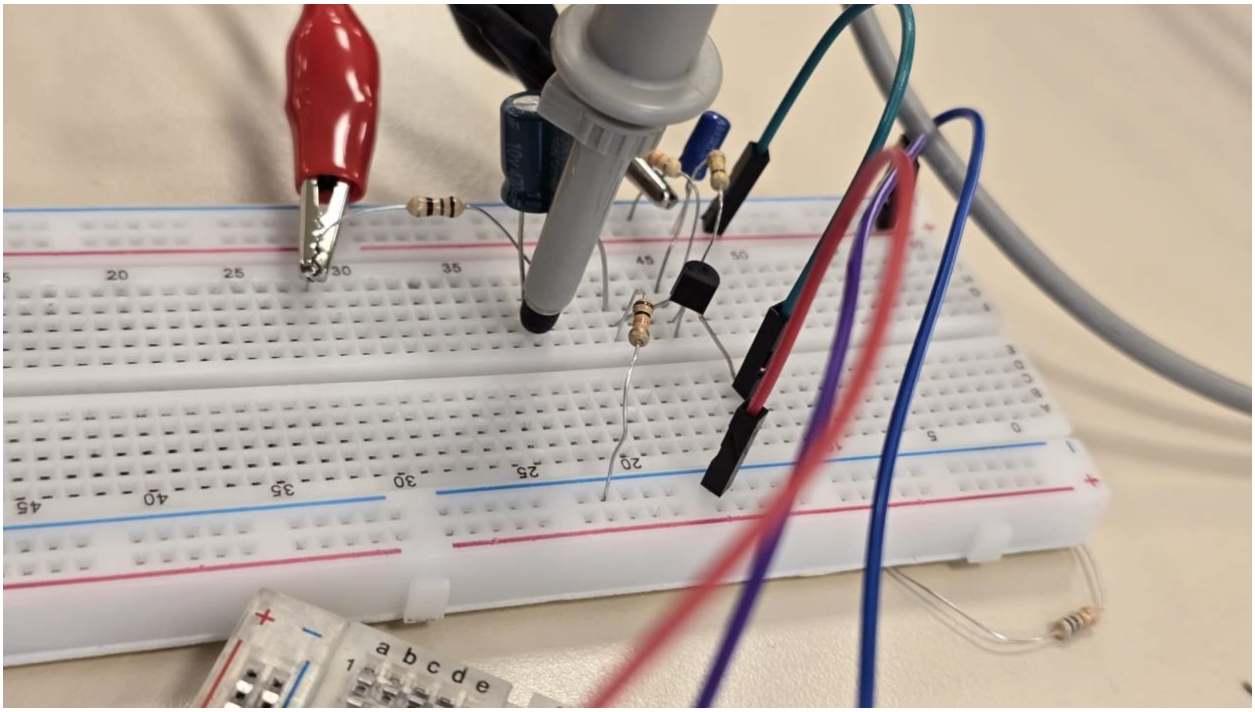


Figure 17. Breadboard implementation of the output stage during in-lab testing.

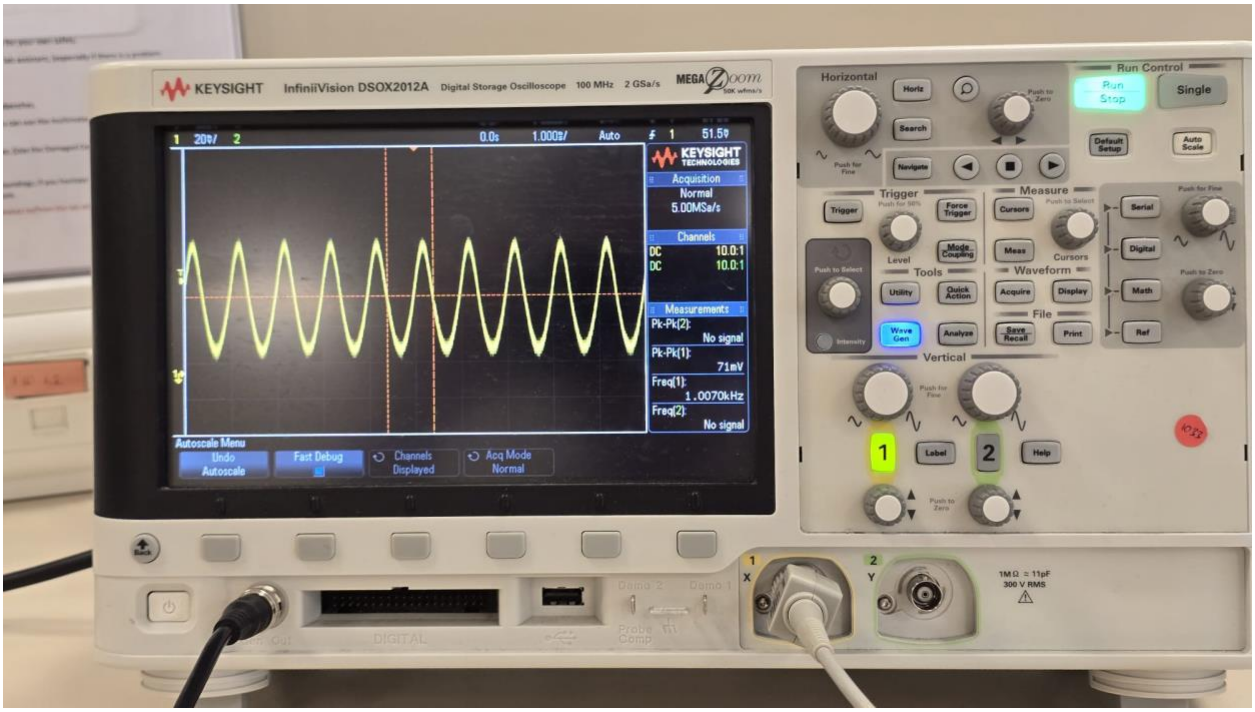


Figure 18. Oscilloscope measurement used for the output impedance test.

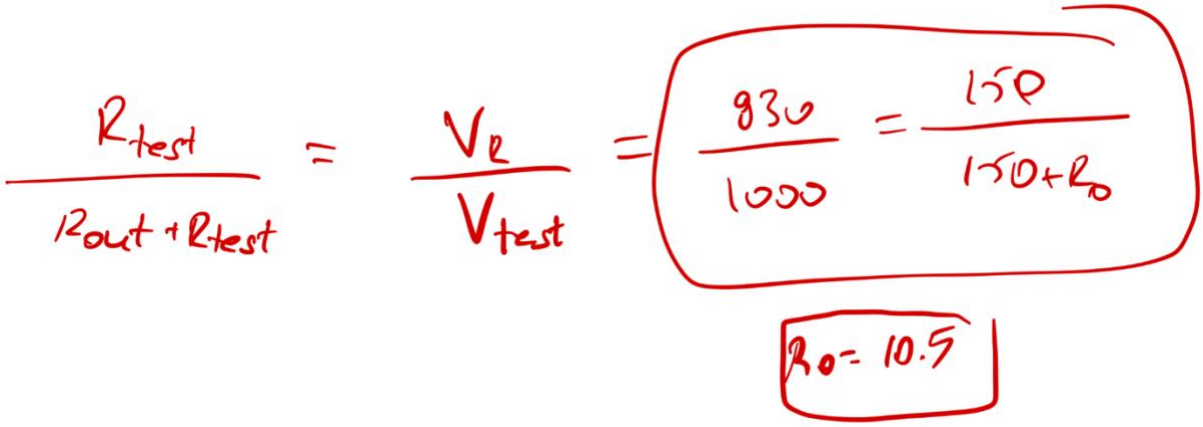


Figure 19. Hand calculation of output resistance using the voltage divider method.

The output stage was built using Q4, a 2N2222 transistor connected in common-collector configuration, also called an emitter follower. This topology was chosen because an emitter follower has voltage gain close to unity, high input impedance, and low output impedance. In the complete amplifier, Q2 and Q3 already provide the required voltage gain, so Q4 does not need to amplify the voltage further. Instead, Q4 buffers the amplified signal and provides enough output current to drive the 200 Ω load. Without this output buffer, the amplification stage would be directly loaded by the small 200 Ω resistance, which would reduce the gain significantly and could cause distortion.

The components used in this stage were Q4 = 2N2222, R5 = 3 MΩ, R12 = 100 kΩ, R13 = 1 kΩ, C5 = 470 µF, and RL = 200 Ω in normal operation. For the output impedance measurement, RL was removed and a 150 Ω test resistor was used instead.

R5 and R12 form the bias network for the base of Q4. R5 is connected from the positive supply side to the base, while R12 pulls the base toward the lower reference node. These resistors set the DC base voltage of Q4. The value of R5 was selected much larger than R12 so that the base voltage would be biased at a level that allows the emitter to sit at a suitable DC voltage and conduct enough current through R13. The output stage needs more current than the previous stages because it must drive the 200 Ω load. In the pre-lab design, Q4 was intentionally biased at a higher current than Q1, Q2, and Q3. This is important because the small-signal emitter resistance of a BJT is approximately re ≈ VT / IE. Therefore, a larger emitter current gives a smaller re, and a smaller re helps reduce the output impedance of the emitter follower.

R13 was selected as 1 kΩ and connected at the emitter side of Q4. This resistor sets the DC emitter current together with the base bias. A relatively high emitter current is necessary because the load current required for a 2.5 Vpp output across 200 Ω is significant. For a 2.5 Vpp output, the peak output voltage is 1.25 V, so the peak load current is IL,peak = 1.25 V / 200 Ω = 6.25 mA. Because of this, Q4 must be biased with enough current to supply the load without entering cutoff or causing distortion. In the pre-lab calculation, the output stage current was designed to be around 10 mA, which is enough to drive the 200 Ω load more linearly.

C5 was selected as 470 µF and used as the output coupling capacitor. Its purpose is to pass the AC output signal to the load while blocking the DC emitter voltage of Q4. This is required because the lab specification states that there should be no DC current through the load. If the load were connected

directly to the emitter without C5, the DC bias current could flow through the 200 Ω load and disturb both the operating point and the load voltage. By placing C5 in series with the load, the AC signal is delivered to RL while the DC component is blocked.

The value of C5 had to be much larger than the other coupling capacitors because it works with the small 200 Ω load resistance. The cutoff frequency of an output coupling capacitor is approximately $f_C = 1 / [2\pi C(R_{out} + R_L)]$. Since RL is only 200 Ω, a small capacitor would create a high cutoff frequency and attenuate the low-frequency signal. Therefore, C5 was chosen as 470 µF to keep the output high-pass pole below the required 20 Hz lower cutoff. In the pre-lab calculation, using C5 = 470 µF with the output resistance and 200 Ω load gave an output coupling pole around 1–2 Hz, which is safely below the specification.

The theoretical output impedance of an emitter follower is much smaller than the emitter resistor alone because the transistor provides current gain. A simplified expression for the output resistance is $R_{out} \approx (r_e + R_{B,source} / (\beta + 1)) \,||\, R13$, where re is the small-signal emitter resistance and $R_{B,source}$ is the resistance seen looking back into the base network and previous stage. Since Q4 is biased at a relatively high emitter current, re becomes small. This is why the emitter follower can provide a low output impedance and drive the load more effectively. In the pre-lab simulation, the output impedance was expected to be well below 100 Ω, around several tens of ohms.

Measurements

To measure the output impedance of the output stage, the normal 200 Ω load resistor was removed. Then a 1 Vpp test signal was applied for the output impedance test, and a 150 Ω test resistor was connected at the output. The reason for removing RL was to measure the internal output resistance of the output stage itself without the 200 Ω load affecting the result. The 150 Ω test resistor was used as a known resistor in a voltage divider with the unknown output resistance Ro of the stage.

During this measurement, the output stage was treated as a Thevenin source with an internal resistance Ro. When the 150 Ω test resistor was connected, the applied test voltage divided between Ro and Rtest. The measured voltage across the test resistor was approximately 936 mVpp while the applied test voltage was 1 Vpp. Therefore, most of the test voltage appeared across the 150 Ω resistor, which indicates that the internal output resistance was much smaller than 150 Ω.

Using the voltage divider relationship, $V_{Rtest} / V_{test} = R_{test} / (R_{test} + R_o)$. Substituting the measured values gives 936 mV / 1000 mV = 150 Ω / (150 Ω + Ro). Solving for Ro gives Ro ≈ 10.5 Ω.

Therefore, the measured output resistance of the output stage was approximately 10.5 Ω. This is much lower than the required maximum output impedance of 100 Ω. This result confirms that the common-collector output buffer successfully provides a low output impedance.
The result also agrees with the expected behavior of an emitter follower. Since Q4 is biased at a relatively high current, its small-signal emitter resistance is low. A low internal emitter resistance allows the output voltage to remain almost constant even when a load is connected. This means the stage can deliver current to the load without the output voltage dropping significantly. This is exactly why the output stage was implemented as a common-collector buffer instead of another common-emitter gain stage.

The measured output impedance was lower than the pre-lab simulated value, which was approximately 47 Ω for the complete amplifier output. This difference is reasonable because the measurement setup in

the lab was not exactly identical to the full LTspice output impedance simulation. In the lab, the output stage was tested with RL removed and with a 150 Ω test resistor added for the voltage divider measurement. The actual transistor β, the operating current of Q4, and the exact base-driving resistance can also differ from the simulated assumptions. If the real Q4 had a higher effective current or if the base was driven from a lower resistance than expected during the test, the measured output resistance could become lower than the pre-lab value.

Another important point is that the output impedance measurement is sensitive to small voltage differences. Since the measured voltage across Rtest was 936 mV out of a 1 V test signal, only about 64 mV was effectively dropped across the output resistance. Small oscilloscope reading errors, probe loading, contact resistance, and generator amplitude uncertainty can noticeably affect the calculated Ro. However, even with these possible errors, the measured value is far below 100 Ω, so the conclusion is still reliable.

There are also practical differences between LTspice and the breadboard circuit. LTspice uses transistor model parameters that may not exactly match the real 2N2222 transistor used in the lab. The real β value can be different from the assumed value, and the transistor's actual operating point can shift due to resistor tolerances and breadboard connections. These effects can change the small-signal emitter resistance and base current, which directly affect the measured output impedance. However, the emitter follower topology is robust against these variations, and the measured result still satisfies the design goal.

In normal operation, the 200 Ω load is connected through C5. Since the measured output resistance is about 10.5 Ω, it is much smaller than the 200 Ω load. This is desirable because it means most of the output voltage appears across the load rather than being lost inside the amplifier. If the output resistance were close to or larger than 200 Ω, the load voltage would drop significantly and the amplifier would not be able to deliver the required output swing. Because Ro is much smaller than RL, the output stage is suitable for driving the load.

In conclusion, the output stage successfully worked as a low-output-impedance buffer. Q4 was used as an emitter follower to provide current gain and drive the load, while R5, R12, and R13 set the DC bias point. C5 blocked DC current from flowing through the load and allowed only the AC output signal to pass. The measured output resistance was approximately 10.5 Ω using the voltage divider method with a 150 Ω test resistor and a 1 Vpp test signal. This value is well below the required 100 Ω limit, so the output stage satisfies the output impedance specification and is suitable for connection to the full multistage amplifier.

3.4 Part 4: Complete Multistage Amplifier

Building Block

The fourth checkpoint of the in-lab assignment was to connect all previously tested stages together and show that the complete multistage amplifier works as desired. After testing the input stage, amplification stage, and output stage separately, the full circuit was assembled on the breadboard. The purpose of this final part was to verify that the complete amplifier satisfies the main design specifications: a non-inverting midband gain close to 100 V/V, lower 3-dB frequency below 20 Hz, upper 3-dB frequency above 20 kHz, high input impedance, low output impedance, and a clean output swing

without visible nonlinear distortion. The lab handout specifically requires the complete circuit to be shown working after the individual stages are built and tested.

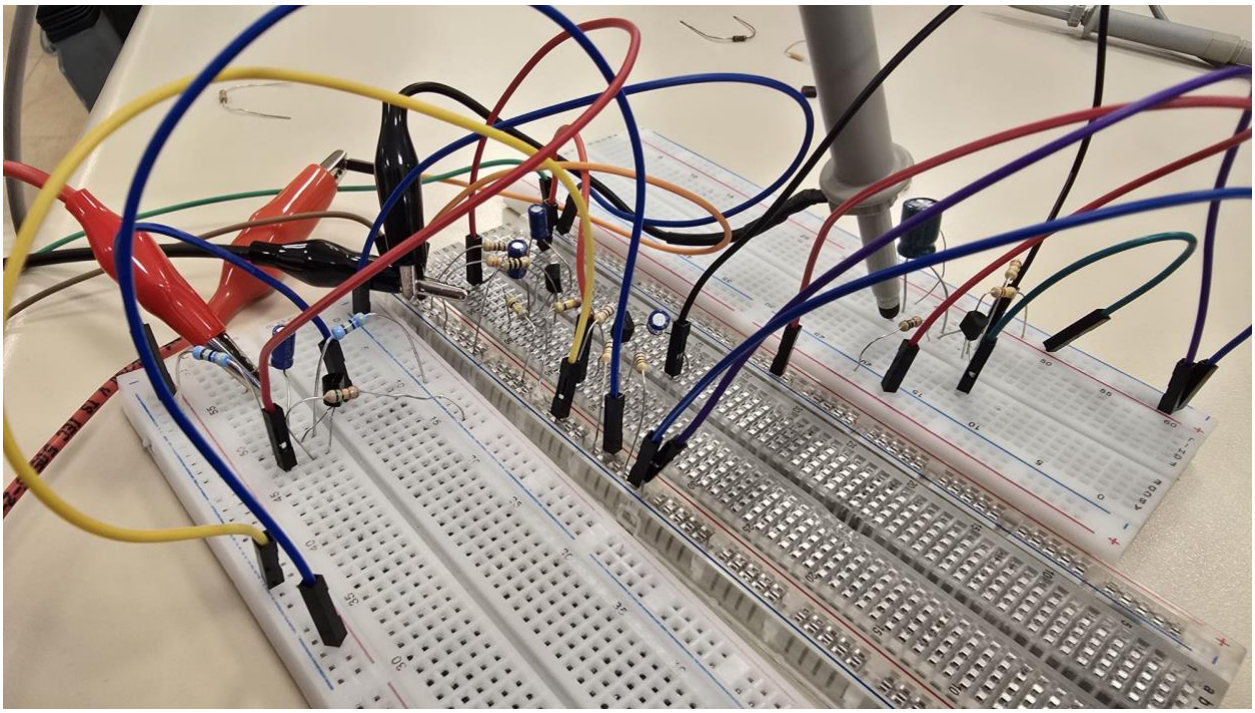


Figure 20. Breadboard implementation of the complete multistage amplifier.

The complete amplifier consists of four BJT stages using 2N2222 transistors. Q1 is the input common-collector stage, Q2 and Q3 are the common-emitter voltage-gain stages, and Q4 is the output common-collector stage. Each stage has a specific role in the overall design. Q1 provides high input impedance so that the source is not loaded. Q2 and Q3 provide most of the voltage gain. Q4 provides low output impedance so that the amplifier can drive the 200 Ω load. This combination of buffer-gain-gain-buffer makes the circuit more practical than using only one or two high-gain stages.

The first stage, Q1, is an emitter follower. Its voltage gain is close to 1, so it does not significantly increase the signal amplitude. However, it is important because it increases the input impedance of the whole amplifier. Without this input buffer, the function generator would directly see the biasing network and input resistance of the common-emitter stage, which could reduce the input impedance below the required value. In the previous input-stage test, the measured input impedance was about 1.47 MΩ, which was higher than the required 1 MΩ. Therefore, Q1 successfully works as the input buffer.

The second and third stages, Q2 and Q3, are common-emitter amplifiers. These two stages provide the main voltage gain. Each common-emitter stage inverts the signal by 180°. Since two common-emitter stages are used, the signal is inverted twice, so the final output of the amplifier is non-inverting with respect to the input. This is important because the lab required the overall voltage gain to be non-inverting. In the pre-lab design, the gain of the two common-emitter stages was estimated to be close to 100 V/V, and the measured full-circuit gain also remained in the acceptable range.

The fourth stage, Q4, is another emitter follower. Its purpose is to lower the output impedance and provide enough current to drive the 200 Ω load. Since the load resistance is small, directly connecting it to the output of the common-emitter amplification stage would heavily load the circuit and reduce the voltage gain. The output buffer prevents this problem by providing current gain and low output resistance. In the previous output-stage test, the measured output resistance was about 10.5 Ω using the voltage divider method, which is far below the required 100 Ω. Therefore, Q4 is suitable for driving the 200 Ω load.

The complete circuit also uses several coupling capacitors. C1 couples the input signal to Q1, C2 couples the input buffer to Q2, C3 couples Q2 to Q3, C4 couples Q3 to Q4, and C5 couples the output of Q4 to the 200 Ω load. These capacitors allow the AC signal to pass from one stage to the next while blocking DC. This is necessary because each transistor stage has its own DC bias point. If the stages were directly connected without coupling capacitors, the DC bias of one stage would disturb the next stage and could move the transistors out of the active region. C5 is especially important because it blocks DC current from flowing through the load, satisfying the requirement that no DC current should flow through the load.

For the complete circuit, the input signal was set to 25 mVpp with 0 V DC offset. This small input amplitude was selected because the expected gain was around 100 V/V. With this gain, a 25 mVpp input should produce an output around 2.5 Vpp. This is close to the required output swing and large enough to observe clearly on the oscilloscope, but still small enough to avoid clipping. If the input signal were much larger, the output could exceed the available linear swing of the BJT stages and become distorted.

Measurements

The complete amplifier was first tested at the midband frequency of approximately 1 kHz. At this frequency, the coupling capacitors do not significantly attenuate the signal, and high-frequency parasitic capacitances are not dominant yet. Therefore, the 1 kHz measurement was used as the midband reference.

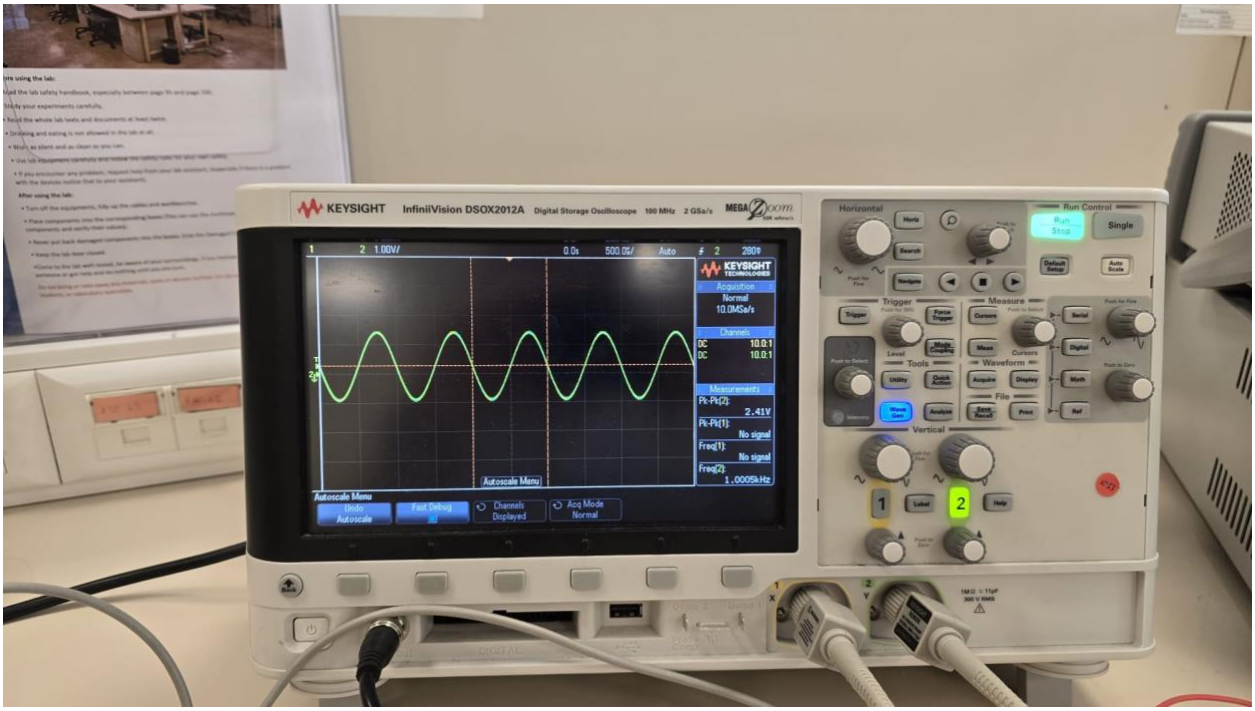


Figure 21. Oscilloscope measurement of the complete amplifier at the 1 kHz midband frequency.

At 1 kHz, the measured output voltage was 2.41 Vpp for a 25 mVpp input. The midband voltage gain was Av = Vout / Vin = 2.41 Vpp / 25 mVpp = 2.41 / 0.025 = 96.4 V/V. This value is within the required gain range of 100 ± 15%, which corresponds to 85 V/V to 115 V/V. The waveform was sinusoidal and did not show visible clipping, which means that the complete amplifier was operating in the linear region. This result also shows that the output buffer did not significantly reduce the voltage gain while allowing the circuit to drive the load.

To determine the 3-dB level, the midband output voltage was divided by √2. Therefore, V3dB = Vmid / √2 = 2.41 Vpp / √2 = 1.70 Vpp. This means that the lower and upper cutoff frequencies occur when the output voltage drops from 2.41 Vpp to approximately 1.70 Vpp.

For the lower-frequency measurement, the frequency was decreased until the output voltage approached the 3-dB level. At approximately 5.79 Hz, the measured output voltage was 1.61 Vpp. The gain at this frequency was Av = Vout / Vin = 1.61 Vpp / 25 mVpp = 64.4 V/V. The ideal 3-dB gain based on the midband gain is Av,3dB = 96.4 / √2 = 68.2 V/V. The measured value is slightly below the exact 3-dB gain, so the exact lower 3-dB frequency is probably slightly higher than 5.79 Hz. However, this measurement clearly shows that the lower cutoff frequency is well below 20 Hz. Therefore, the full amplifier satisfies the lower bandwidth requirement.

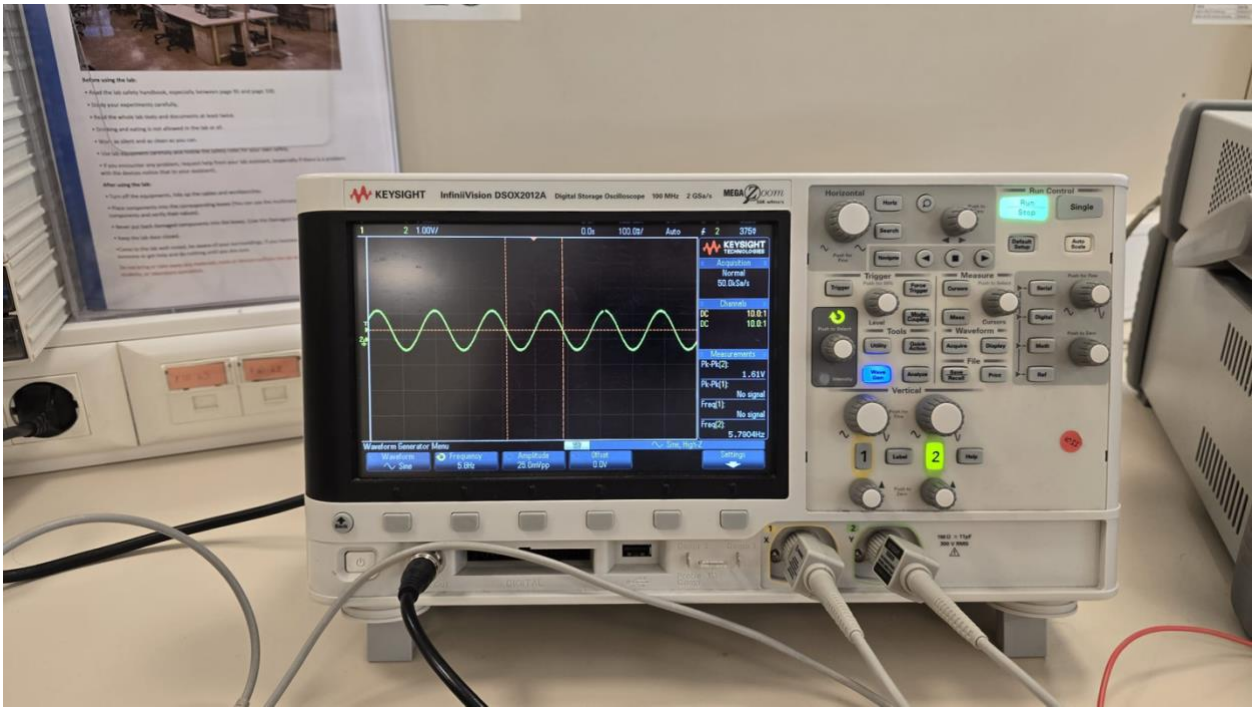


Figure 22. Oscilloscope measurement of the complete amplifier near the lower 3-dB frequency.

For the upper-frequency measurement, the frequency was increased until the output voltage dropped close to the calculated 3-dB value. At approximately 340 kHz, the measured output voltage was about 1.73 Vpp. The gain at this frequency was Av = Vout / Vin = 1.73 Vpp / 25 mVpp = 69.2 V/V. This is very close to the calculated 3-dB gain of 68.2 V/V. Therefore, the upper 3-dB frequency of the complete amplifier can be estimated to be around 340 kHz. Since this is much higher than the required 20 kHz, the complete amplifier satisfies the upper bandwidth requirement.

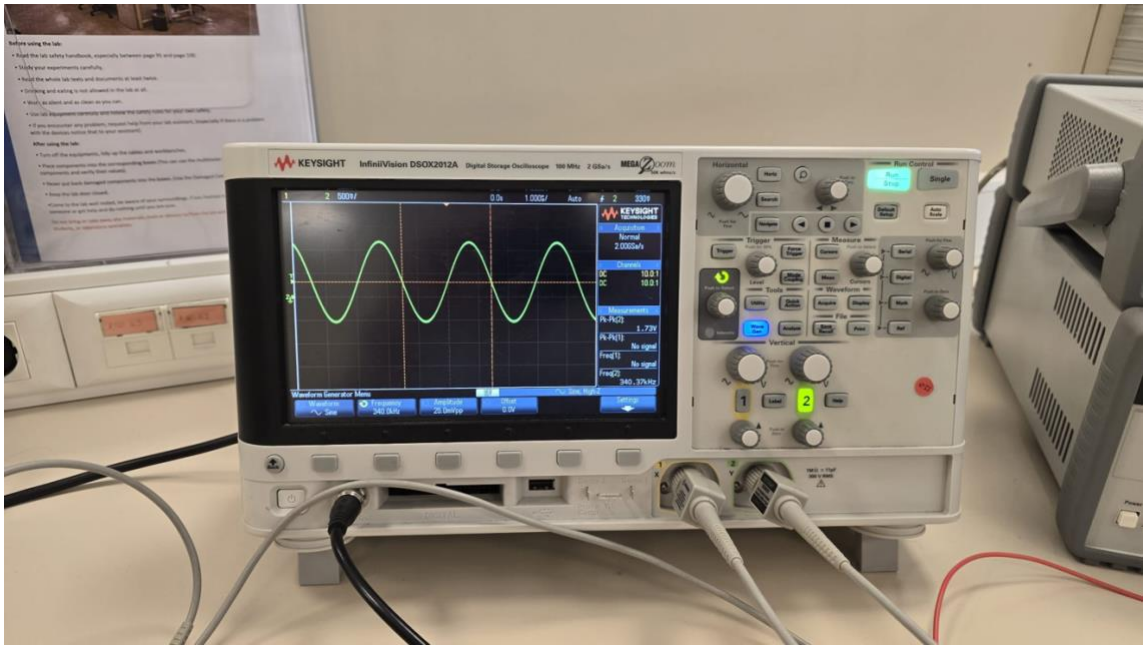


Figure 23. Oscilloscope measurement of the complete amplifier near the upper 3-dB frequency.

The measured values can be summarized as follows: at the lower cutoff region, f ≈ 5.79 Hz, Vin = 25 mVpp, Vout = 1.61 Vpp, and Av ≈ 64.4 V/V; at the midband frequency, f ≈ 1.0005 kHz, Vin = 25 mVpp, Vout = 2.41 Vpp, and Av ≈ 96.4 V/V; at the upper cutoff region, f ≈ 340 kHz, Vin = 25 mVpp, Vout = 1.73 Vpp, and Av ≈ 69.2 V/V. Since the calculated 3-dB output voltage was approximately 1.70 Vpp, both cutoff-region measurements are close to the expected 3-dB level.

The complete circuit also satisfied the impedance-related requirements based on the previous stage tests. The input stage measurement gave Zin ≈ 1.47 MΩ, which is greater than the required 1 MΩ. The output stage measurement gave Ro ≈ 10.5 Ω, which is lower than the required 100 Ω. Therefore, when all stages are connected, the circuit has the correct buffering behavior at both the input and output.
Comparison with Theory and Pre-Lab Results

The measured midband gain of the complete amplifier was 96.4 V/V, while the pre-lab theoretical and LTspice result was around 106–107 V/V. The measured value is slightly lower than the simulation, but it still satisfies the required range. This difference is expected because the real circuit was built on a breadboard using actual 2N2222 transistors and standard-value components.

One major reason for the difference is the variation of the transistor β values. In the pre-lab calculations, β was assumed to be approximately 200. In the lab, the actual 2N2222 transistors can have lower β values, and the β values of Q1, Q2, Q3, and Q4 may not be the same. This changes base currents, bias voltages, emitter currents, and loading between stages. Since the voltage gain and impedance of each stage depend on the transistor operating point, the measured full-circuit result can differ from the LTspice result.

Another reason is stage loading. In the complete circuit, each stage loads the previous stage. Q2 loads the output of Q1, Q3 loads the collector of Q2, and Q4 loads the collector of Q3. The output load also affects the final stage through C5. These loading effects were included approximately in the pre-lab calculation, but the actual loading in the breadboard circuit can be different due to real transistor β, resistor tolerances, and probe loading. Since the total gain is the product of the gains of multiple stages, small differences in individual stages can produce a noticeable difference in the final gain.

The lower 3-dB result is mainly caused by the coupling capacitors. At low frequencies, the reactance of C1, C2, C3, C4, and C5 increases. This reduces signal transfer between stages and causes the output amplitude to drop. C5 is especially important because it works with the small 200 Ω load, so it was chosen as 470 µF to keep the output high-pass pole low. The measured lower cutoff region around 5.79 Hz confirms that the coupling capacitors were large enough because the lower cutoff is below the required 20 Hz.

The upper 3-dB result is mainly caused by transistor parasitic capacitances, oscilloscope probe capacitance, breadboard capacitance, and the Miller effect in the common-emitter stages. At high frequency, the base-collector capacitance of Q2 and Q3 is effectively increased by the gain of each stage. This creates high-frequency poles and reduces gain. The measured upper cutoff around 340 kHz is

consistent with this behavior. Since this frequency is far above 20 kHz, the high-frequency performance of the complete amplifier is more than sufficient for the lab requirement.

The waveform quality is also important. At the midband test frequency, the output waveform was sinusoidal and did not show visible nonlinear distortion. This means that the transistors were biased properly and that the selected input amplitude did not push the amplifier into clipping. If the bias points were incorrect or if the input signal were too large, the output waveform would have clipped at the top or bottom. Since the waveform remained clean, the DC bias design and emitter degeneration were effective.

The complete multistage amplifier worked as intended. The measured midband output was 2.41 Vpp for a 25 mVpp input, giving a gain of approximately 96.4 V/V, which is within the required 100 ± 15% range. The lower cutoff region was measured around 5.79 Hz, which is below the required 20 Hz limit, and the upper cutoff region was measured around 340 kHz, which is far above the required 20 kHz limit. The previously measured input and output impedances also satisfied the requirements, with Zin above 1 MΩ and Ro below 100 Ω. Therefore, the complete circuit successfully met the gain, bandwidth, input impedance, output impedance, and output swing specifications.

4. Discussion/Analysis

The purpose of this lab was to design, simulate, build, and test a discrete multistage BJT amplifier that satisfies the required gain, bandwidth, input impedance, output impedance, and output swing specifications. The amplifier was designed using four 2N2222 BJT stages: an input common-collector buffer, two common-emitter voltage-gain stages, and an output common-collector buffer. This structure was selected because a single BJT stage cannot easily satisfy all of the requirements at the same time. The common-emitter stages provide the necessary voltage gain, while the common-collector stages improve the input and output impedance characteristics. According to the lab specifications, the amplifier should have a non-inverting midband gain of 100 ± 15%, a lower 3-dB frequency below 20 Hz, an upper 3-dB frequency above 20 kHz, an input impedance higher than 1 MΩ, an output impedance lower than 100 Ω, and a clean 2.5 Vpp output swing at 1 kHz.

The circuit was first designed and analyzed in the pre-lab using hand calculations and LTspice simulations. The pre-lab result predicted a midband gain of approximately 106–107 V/V, an input impedance around 1.72–1.8 MΩ, and an output impedance around 47–50 Ω. These values satisfied the original design requirements. In the lab, the stages were tested separately first, as recommended in the lab handout, and then the complete amplifier was connected and tested. This stage-by-stage method was useful because it allowed each part of the circuit to be verified before combining everything into the complete system.

The comparison between the theoretical, simulation, and measured results is summarized below:

Parameter	Required value	Pre-lab / LTspice result	Measured result	Evaluation
Midband gain	85–115 V/V	106–107 V/V	96.4 V/V	Passed
Lower 3-dB frequency	≤ 20 Hz	Below 20 Hz	≈ 5.8 Hz	Passed
Upper 3-dB frequency	≥ 20 kHz	Far above 20 kHz	≈ 340 kHz	Passed
Input impedance	≥ 1 MΩ	≈ 1.72–1.8 MΩ	≈ 1.47 MΩ	Passed
Output impedance	≤ 100 Ω	≈ 47–50 Ω	≈ 10.5 Ω	Passed
Output swing	2.5 Vpp without distortion	≈ 2.7 Vpp	≈ 2.41 Vpp at 1 kHz	Passed

Table 1.

The measured midband gain of the complete amplifier was approximately 96.4 V/V, obtained from a 25 mVpp input and a 2.41 Vpp output at 1 kHz. This value is lower than the pre-lab prediction of about 106–107 V/V, but it is still inside the required range of 85–115 V/V. Therefore, the gain requirement was satisfied. The difference between the simulated and measured gain is reasonable because the real circuit was built with physical 2N2222 transistors, breadboard connections, and standard resistor/capacitor values. In contrast, LTspice uses transistor models and ideal connections that do not perfectly match the actual lab setup.

One of the most important reasons for the gain difference is the variation of the transistor current gain, β. In the pre-lab hand calculations, β was assumed to be approximately 200. However, the actual β of the 2N2222 transistors used in the lab can be much lower and can vary significantly from transistor to transistor. This affects the base current, the loading of the bias dividers, the emitter current, and the small-signal parameters. Since the common-emitter gain depends on the operating point and the loading between stages, a different real β value can shift the measured gain away from the theoretical result.

The design reduced this sensitivity by using emitter degeneration resistors in the common-emitter stages. R9 and R11 were selected as 820 Ω and were not bypassed by capacitors. This makes the voltage gain depend more on external resistor values than directly on transistor β. The approximate gain expression for each common-emitter stage is $A_v \approx -R_{C,eff} / (R_E + r_e)$. Since R_E is much larger than the internal small-signal emitter resistance r_e, the gain becomes more stable and predictable. However, emitter degeneration cannot completely remove the effects of β variation because β still affects input resistance, base current, stage loading, and the DC operating point.

Component tolerances also contributed to differences between theory and measurement. The theoretical calculations used nominal resistor and capacitor values, such as R2 = R4 = 12 kΩ and R9 = R11 = 820 Ω. In the real circuit, each resistor can be slightly different from its labeled value. If the collector resistors are slightly lower than expected, the gain decreases. If the emitter resistors are slightly higher

than expected, the gain also decreases. Since Q2 and Q3 are cascaded, a small gain difference in each stage can multiply and create a noticeable change in total gain.

The measured lower 3-dB frequency was around 5.8 Hz, which is well below the required limit of 20 Hz. This result shows that the coupling capacitors were selected properly. At low frequencies, the reactance of the coupling capacitors increases, which reduces the signal transferred between stages. In this amplifier, C1, C2, C3, and C4 were selected as 1 µF, while C5 was selected as 470 µF. C5 had to be much larger because it works with the low 200 Ω load resistance. If C5 had been too small, the output coupling pole would have moved to a higher frequency and the low-frequency response would not have satisfied the requirement. The measured lower cutoff confirms that the capacitor values were large enough for the required bandwidth.

The measured upper 3-dB frequency was around 340 kHz, which is much higher than the required 20 kHz. This means the amplifier maintains its gain over the required frequency range and only begins to roll off at much higher frequencies. The high-frequency gain drop is mainly caused by transistor parasitic capacitances, oscilloscope probe capacitance, breadboard parasitic capacitance, and the Miller effect in the common-emitter stages. In common-emitter amplifiers, the base-collector capacitance is effectively multiplied by the gain of the stage, increasing the effective input capacitance and creating high-frequency poles. Since the gain was divided between two moderate-gain stages rather than one very high-gain stage, the Miller effect was less severe and the upper cutoff frequency remained high.

The input impedance measurement also satisfied the specification. The measured input impedance was approximately 1.47 MΩ, compared with the pre-lab prediction of about 1.72–1.8 MΩ. Although the measured value was lower, it was still above the required 1 MΩ. The lower measured value can again be explained by real transistor β values and practical loading effects. The input stage was tested separately with a representative 27 kΩ load at its output. Since the input resistance of an emitter follower depends on the effective emitter resistance reflected to the base by approximately β + 1, any decrease in emitter-side resistance or β reduces the measured input impedance. Breadboard leakage, oscilloscope probe loading, and the very large 5 MΩ bias resistors also make the input impedance measurement more sensitive to practical nonidealities.

The output impedance measurement was also successful. The measured output resistance was approximately 10.5 Ω using the voltage divider method with a 150 Ω test resistor. This value is far below the required 100 Ω. The output stage used Q4 as a common-collector buffer, which is appropriate because an emitter follower provides low output impedance and current gain. This is important because the amplifier must drive a 200 Ω load. If the output impedance were high, a significant part of the output voltage would be lost inside the amplifier, and the output swing across the load would decrease. Since the measured output resistance was much smaller than 200 Ω, most of the output voltage can be delivered to the load.

The output impedance measured in the lab was lower than the pre-lab estimate of around 47–50 Ω. This difference may be due to the exact output impedance test setup, the actual Q4 operating current, and the base-driving resistance during the measurement. Output impedance measurements are also sensitive to small voltage differences. In the test, the measured voltage across the 150 Ω resistor was very close to the 1 V test voltage, meaning only a small voltage was dropped across the internal output resistance. Because of this, small oscilloscope reading errors or connection resistances can noticeably

affect the calculated value. However, the measured value is still clearly below 100 Ω, so the output impedance requirement is safely satisfied.

The complete amplifier output waveform was sinusoidal at 1 kHz and did not show visible clipping. This indicates that the DC bias points of the transistor stages were appropriate and that the input amplitude was not too large. The input signal was kept at 25 mVpp because the total gain was close to 100 V/V. If the input had been increased too much, the output voltage would have become too large and the common-emitter stages could have entered cutoff or saturation. This would cause clipping at the top or bottom of the waveform. Since the measured output was clean, the selected biasing and signal amplitude were suitable.

Overall, the experimental results were consistent with the theoretical expectations and LTspice simulations. The measured gain was slightly lower than the simulated value, but it remained inside the required tolerance range. The lower and upper cutoff frequencies both satisfied the bandwidth specifications. The input and output impedance measurements also met the requirements. The small differences between theory, simulation, and measurement were mainly caused by real transistor β variation, component tolerances, stage loading, breadboard parasitics, oscilloscope probe effects, and the limitations of the measurement setup.

In conclusion, the multistage BJT amplifier successfully met the design objectives. The input buffer provided high input impedance, the two common-emitter stages provided the required voltage gain, and the output buffer provided low output impedance for driving the 200 Ω load. The final measured gain was approximately 96.4 V/V, the lower cutoff frequency was around 5.8 Hz, the upper cutoff frequency was around 340 kHz, the input impedance was approximately 1.47 MΩ, and the output impedance was approximately 10.5 Ω. These results show that the amplifier satisfies the gain, bandwidth, input impedance, output impedance, and output swing requirements.

5. Post-Lab

In this lab, a discrete multistage BJT amplifier was designed and tested to meet the required gain, bandwidth, input impedance, output impedance, and output swing specifications. The final circuit was built with four 2N2222 transistors and separated into functional blocks: an input buffer, two voltage-gain stages, and an output buffer. The purpose of using multiple stages was to distribute the amplifier requirements between different transistor configurations instead of forcing one stage to satisfy all specifications. The input and output stages were designed as common-collector stages because they are useful for impedance buffering, while the middle two stages were designed as common-emitter stages because they provide voltage amplification.

The design process started with the required specifications. The amplifier had to produce a non-inverting midband voltage gain of 100 ± 15%, which means the acceptable gain range was 85 V/V to 115 V/V. It also had to have a lower 3-dB frequency below 20 Hz and an upper 3-dB frequency above 20 kHz. In addition, the input impedance had to be at least 1 MΩ, the output impedance had to be below 100 Ω, and the amplifier had to drive a 200 Ω load with approximately 2.5 Vpp output at 1 kHz without nonlinear distortion. These requirements determined the choice of each stage.

The first design decision was to place a common-collector stage at the input. This stage, Q1, was used to increase the input impedance of the overall amplifier. The voltage gain of this stage is close to unity, so it does not significantly amplify the input signal. However, it prevents the signal source from being loaded by the following common-emitter stages. Large bias resistors, R1 = 5 MΩ and R6 = 5 MΩ, were selected to keep the input resistance high. In the pre-lab simulation and calculations, the input impedance was expected to be around 1.72–1.8 MΩ. In the lab, the measured input impedance was approximately 1.47 MΩ. Although this was lower than the simulated value, it was still higher than the required 1 MΩ, so the input impedance requirement was satisfied.

The next part of the design was the voltage-gain section. Q2 and Q3 were both common-emitter stages. This configuration was chosen because common-emitter amplifiers can provide significant voltage gain. Two stages were used instead of one because obtaining a gain close to 100 V/V from a single transistor stage would make the circuit more sensitive to transistor parameters and would reduce bandwidth. By dividing the gain between two stages, each stage only needed a moderate gain. Also, since each common-emitter stage inverts the signal once, using two of them makes the total response non-inverting. This was necessary because the specification required a non-inverting amplifier.

The gain stages used 12 kΩ collector resistors and 820 Ω emitter resistors. The collector resistors were selected to provide enough voltage gain, while the emitter resistors provided emitter degeneration. The emitter degeneration is important because it improves stability and linearity. It also makes the gain less dependent on the transistor's beta value. In the pre-lab, the gain of the amplifier was predicted to be around 106–107 V/V. In the experiment, the measured midband gain was approximately 96.4 V/V, calculated from 2.41 Vpp output divided by 25 mVpp input. This value was lower than the LTspice result but still within the required range of 85–115 V/V.

The difference between the simulated gain and the measured gain is mainly related to the real beta values of the 2N2222 transistors. In LTspice and hand calculations, the transistor behavior is based on a model, and in the pre-lab calculation beta was assumed to be around 200. In the actual lab, the beta of each 2N2222 transistor can be much lower and can vary from one transistor to another. This is very important for a multistage BJT amplifier because beta affects base current, bias loading, input resistance, and stage-to-stage loading. If beta is lower than expected, the base current becomes larger. This can pull the base voltage away from the calculated value and change the emitter current. Since the small-signal parameters such as gm and re depend on collector current, the voltage gain also changes.

Beta variation also affects the loading between stages. For example, the collector of Q2 is loaded by the input resistance of Q3. If the actual beta of Q3 is lower than expected, the input resistance of Q3 becomes lower. This makes Q3 load Q2 more heavily, reducing the effective collector resistance of Q2 and therefore reducing Q2's gain. A similar effect can happen between other stages. Since the total gain is the product of the individual gains, a small decrease in each stage can produce a noticeable decrease in the total gain. This explains why the measured gain was 96.4 V/V instead of the simulated 106–107 V/V.

The output stage was designed with Q4 as another common-collector stage. The purpose of this stage was to reduce the output impedance and allow the amplifier to drive the 200 Ω load. The output stage does not provide significant voltage gain, but it provides current gain. This is necessary because the load resistance is low compared with the internal resistance of the voltage-gain stages. Without Q4, the 200 Ω

load would heavily load the previous stage and reduce the output voltage. In the output impedance test, the load was removed and a 150 Ω test resistor was used with a 1 Vpp test signal. Using the voltage divider method, the output resistance was calculated as approximately 10.5 Ω. This value is much lower than the required 100 Ω, so the output impedance requirement was clearly satisfied.

The output coupling capacitor C5 was chosen as 470 µF because it works with the 200 Ω load. This capacitor blocks DC current from flowing through the load while allowing the AC output signal to pass. Since the load resistance is small, a small coupling capacitor would create a high lower cutoff frequency. For this reason, the output capacitor had to be much larger than the other coupling capacitors. The measured lower 3-dB frequency was around 5.8 Hz, which confirms that the coupling capacitors were large enough to keep the low-frequency cutoff below 20 Hz.

The bandwidth measurements also showed that the circuit met the requirements. At 1 kHz, the midband output was 2.41 Vpp. The 3-dB output level was calculated as 2.41 / √2 = 1.70 Vpp. At the low-frequency side, the output dropped close to this level around 5.8 Hz, so the lower 3-dB frequency was below the required 20 Hz. At the high-frequency side, the output dropped to about 1.73 Vpp around 340 kHz, which is close to the calculated 3-dB level. Therefore, the upper 3-dB frequency was much higher than the required 20 kHz.

The high-frequency roll-off is mainly caused by nonideal transistor capacitances and the Miller effect. In common-emitter stages, the base-collector capacitance appears larger because it is multiplied by the voltage gain of the stage. This creates high-frequency poles and reduces the gain at high frequencies. Breadboard wiring and oscilloscope probes also add small parasitic capacitances. These effects are not very visible at 1 kHz, but they become more important at hundreds of kilohertz. This is why the measured gain remained stable through the required frequency range but started to decrease near 340 kHz.

The measured output swing at 1 kHz was approximately 2.41 Vpp for a 25 mVpp input. This was close to the required 2.5 Vpp output swing and the waveform did not show visible clipping. The absence of clipping indicates that the DC operating points of the transistors were suitable and that the input signal was not too large. If the input amplitude had been increased too much, the common-emitter stages could have entered cutoff or saturation, causing nonlinear distortion. Therefore, the selected input amplitude was appropriate for linear operation.

Several practical factors explain why the experimental results were not exactly equal to the LTspice results. The first and most important factor is the difference between the simulated transistor model and the real 2N2222 transistors, especially the beta value. The second factor is component tolerance, since real resistors and capacitors do not have exactly their nominal values. The third factor is breadboard parasitics, including contact resistance, wire resistance, and stray capacitance. The fourth factor is measurement uncertainty from oscilloscope probes, generator settings, and reading small voltage differences during impedance measurements. These effects are expected in a real BJT amplifier and explain the differences between hand calculations, LTspice, and the lab results.

The final experimental results show that the amplifier met all required specifications. The measured midband gain was 96.4 V/V, which is inside the required 85–115 V/V range. The lower 3-dB frequency was around 5.8 Hz, which is below 20 Hz. The upper 3-dB frequency was around 340 kHz, which is above 20 kHz. The input impedance was approximately 1.47 MΩ, which is above 1 MΩ. The output impedance

was approximately 10.5 Ω, which is below 100 Ω. The output waveform at 1 kHz was sinusoidal and close to the required 2.5 Vpp output swing. Therefore, the design, simulation, and experimental implementation were successful, even though the measured gain was slightly lower than the LTspice result due mainly to real transistor beta variation and practical nonidealities.

6. Conclusion

In this experiment, a discrete multistage BJT amplifier was designed, simulated, implemented, and tested successfully. The main objective was to build an amplifier that could provide a non-inverting midband voltage gain close to 100 V/V while also satisfying the required bandwidth, input impedance, output impedance, and output swing specifications. The final circuit used four 2N2222 transistor stages: an input common-collector buffer, two common-emitter voltage-gain stages, and an output common-collector buffer. This structure allowed each part of the circuit to perform a specific function instead of forcing one stage to satisfy all design requirements.

The experimental results showed that the amplifier met the required specifications. The measured midband gain was approximately 96.4 V/V, which is within the required 100 ± 15% range. The lower 3-dB frequency was measured around 5.8 Hz, which is below the required 20 Hz limit, and the upper 3-dB frequency was measured around 340 kHz, which is far above the required 20 kHz limit. The measured input impedance was approximately 1.47 MΩ, satisfying the requirement of at least 1 MΩ, and the measured output impedance was approximately 10.5 Ω, satisfying the requirement of less than 100 Ω. The output waveform at 1 kHz was sinusoidal and close to the expected 2.5 Vpp output swing, showing that the amplifier operated without visible nonlinear distortion.

The results support the design hypothesis that a multistage amplifier using buffer stages and common-emitter gain stages can meet practical amplifier specifications more effectively than a single-stage design. The input common-collector stage improved the input impedance, the two common-emitter stages supplied the required voltage gain, and the output common-collector stage reduced the output impedance and allowed the circuit to drive the 200 Ω load. Therefore, the chosen stage arrangement was valid and appropriate for the required performance.

Although the measured results were not exactly the same as the LTspice simulations, the differences were expected. The most important reason was the difference between the assumed transistor beta value in the pre-lab and the actual beta values of the real 2N2222 transistors used in the lab. Since beta affects base current, biasing, input resistance, loading between stages, and gain, variations in real transistor beta caused the measured gain to be slightly lower than the simulated value. Component tolerances, breadboard parasitics, oscilloscope probe loading, and small measurement uncertainties also contributed to the differences between theory, simulation, and experiment.

Overall, the experiment demonstrated the importance of stage-by-stage amplifier design and testing. By separately verifying the input buffer, amplification stages, output buffer, and then the complete circuit, it was easier to identify whether each block satisfied its function. The final amplifier successfully achieved

the required gain, bandwidth, input impedance, output impedance, and output swing specifications. Therefore, the design procedure, simulation work, and experimental implementation were successful.